



ECON 100B: Macroeconomics

Class 20: Stabilization policy [1/2]
(Jones Chapter 13)

Ryan D. Edwards
April 9, 2026

Agenda

- Finish last vignette of Chapter 12 on the [Great Inflation](#) of the 1970s and the [Great Moderation](#)
- On to Chapter 13 and monetary policy **rules** and stabilization policy

Same announcements

- **Problem Set 3** is available, due on **Friday, April 17**
- **Problem Set 4** will be due on **Friday, May 1**
- The **Final exam** is confirmed for **Monday, May 11**
 - Just like at midterm, the main class meets here in **Pimentel 1**
 - We will run 90 minutes from about 3:00-4:30PM
 - DSP students: Please either make a proctoring appointment, or for 150% time only, we will have **Anthropology and Art Practice 160**
(I think it's kind of near Caffe Strada on Bancroft)

Current events

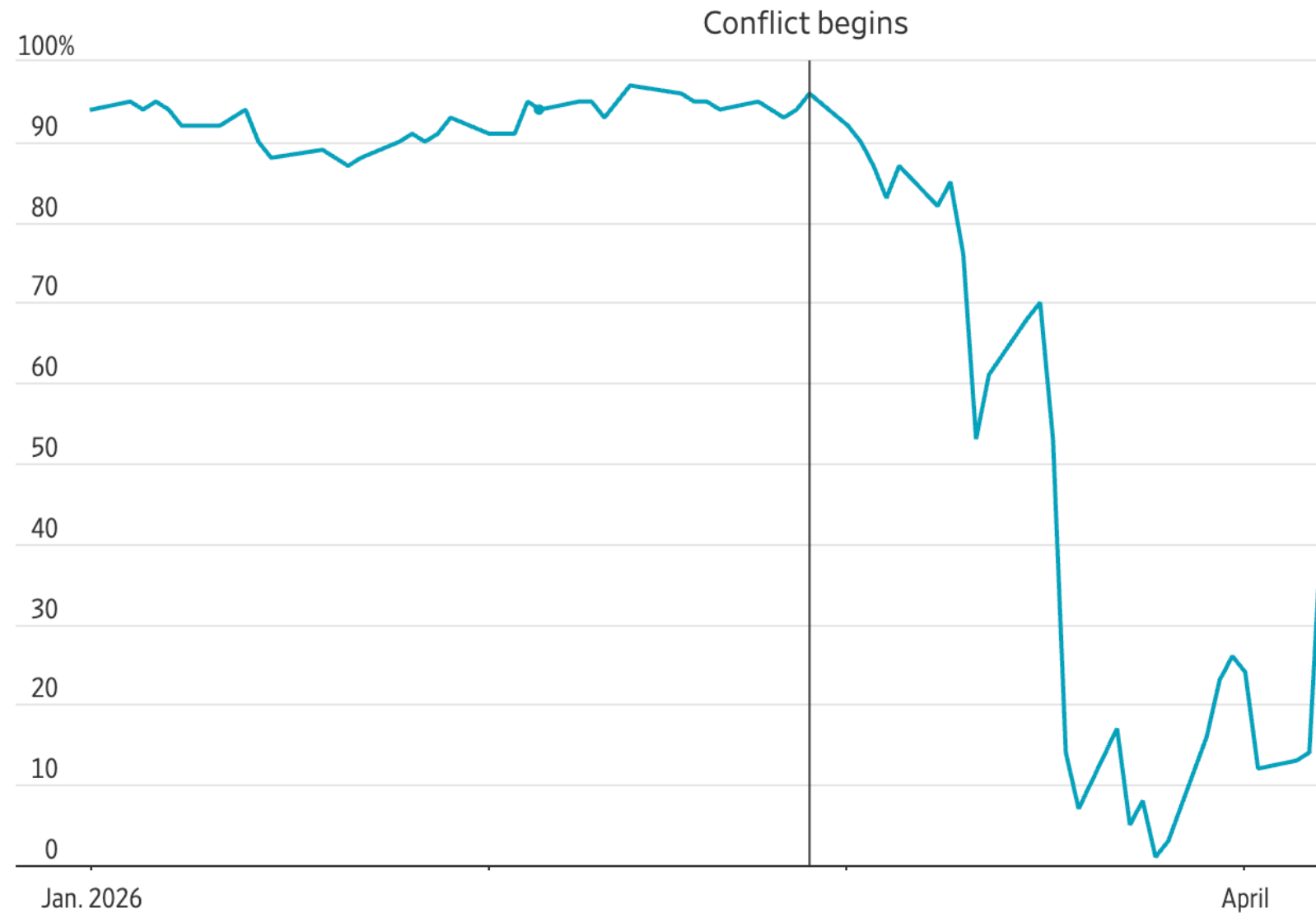
• LIVE 18 hours ago

A Fed Interest-Rate Cut Is Back in Play [Link to article](#)



By Chelsey Dulaney, Reporter

Probability that the Fed cuts rates in 2026

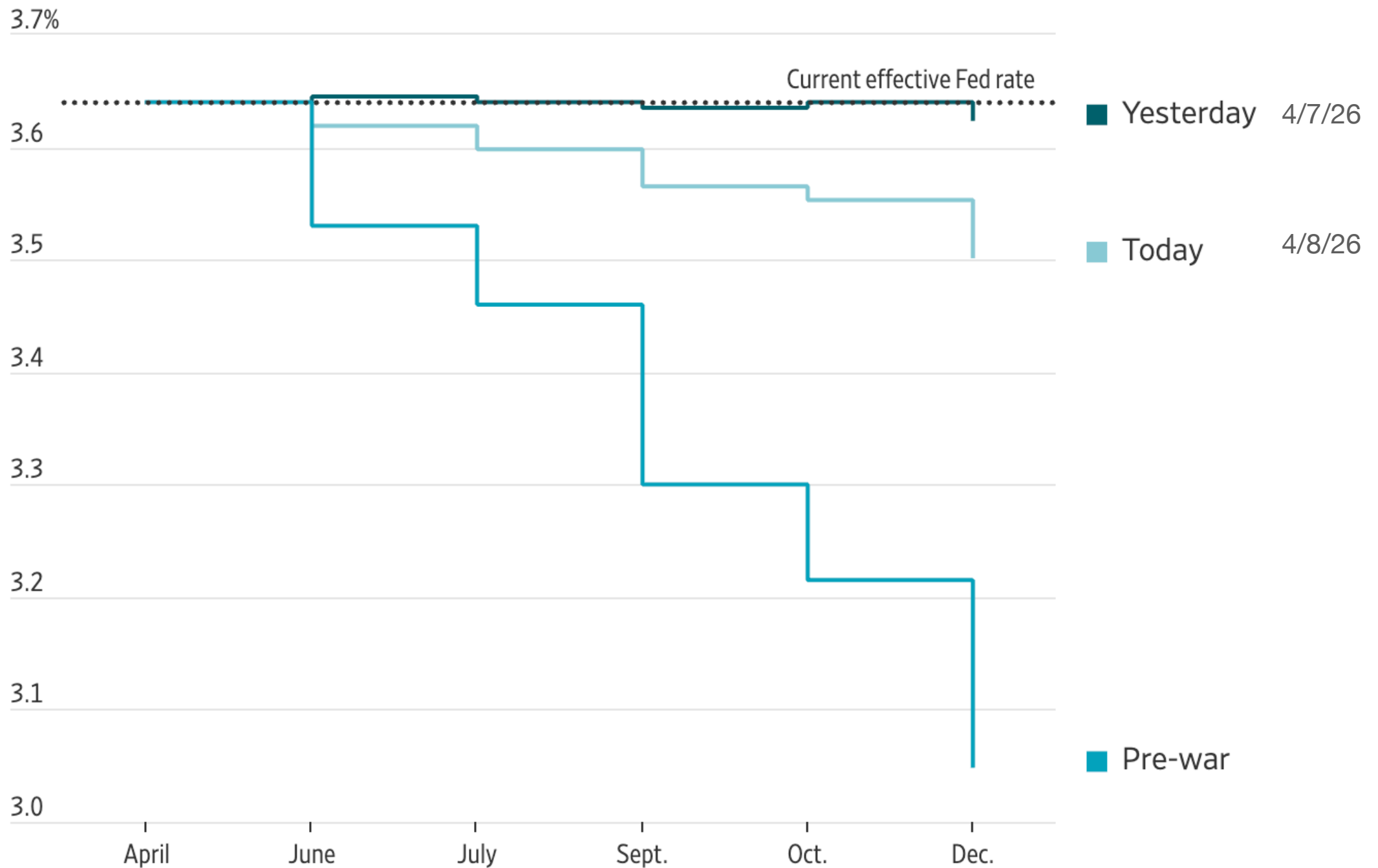


Note: Derived from federal-funds futures contracts
Source: CME Group

Current events

Fed's expected rate path in 2026

[Link to article](#)



Note: Derived from federal-funds futures contracts. Pre-war rate expectations are from Feb. 27.

Source: LSEG

Where we are and where we're headed

Chapter 9: The Short Run

Short-run output $\tilde{Y} = (Y - \bar{Y})/\bar{Y}$

Chapter 10: Historical Macro-clysms

$MV = PY$ and when M collapses, $\pi < 0 \rightarrow R > i$

Chapter 11: The IS Curve

$\tilde{Y} = \bar{a} - \bar{b}(R - \bar{r})$ because I falls when R rises

Chapter 12: Monetary Policy

The central bank chooses R in the short run

and the Phillips Curve

$\Delta\pi = \bar{v} \tilde{Y}$ because inflation increases when $\tilde{Y} > 0$

Chapter 13: Stabilization Policy

Central banks have an inflation target like $\bar{\pi} = 0.02$

Chapter 14: Financial Crises

Like deflation, risk premiums can also push R too high

Chapter 18: Government

Government can stabilize demand but can't spend forever

Chapter 19: International Trade

Trade deficits are not all they seem; $S - I = NX$

Chapter 20: Exchange Rates

Rising R attracts foreign money, causes appreciation, $NX \downarrow$

Learning objectives for Chapter 13

- When **shocks** hit aggregate demand or energy supplies and prices, governments often try to stabilize their economies
 - **Monetary policy** adjusts the short-term interest rate in a currency area like the U.S., the Euro Area, India, China, etc.
 - **Fiscal policy** adjusts spending G to directly change Y or adjusts taxes T and transfer payments Tr to indirectly change Y (Chapter 18)
- Adjusting policy is important to do in good times, when $\tilde{Y}$ and $\Delta\pi$ are too high, and in bad times, when $\tilde{Y}$ and $\Delta\pi$ are too low
- Central banks in advanced countries often set monetary policy to follow a **rule** based around an **inflation target** usually $\bar{\pi} = 0.02$
- By following the rule, central banks **build credibility** and anchor agents' inflation expectations, so that they actually can manage inflation

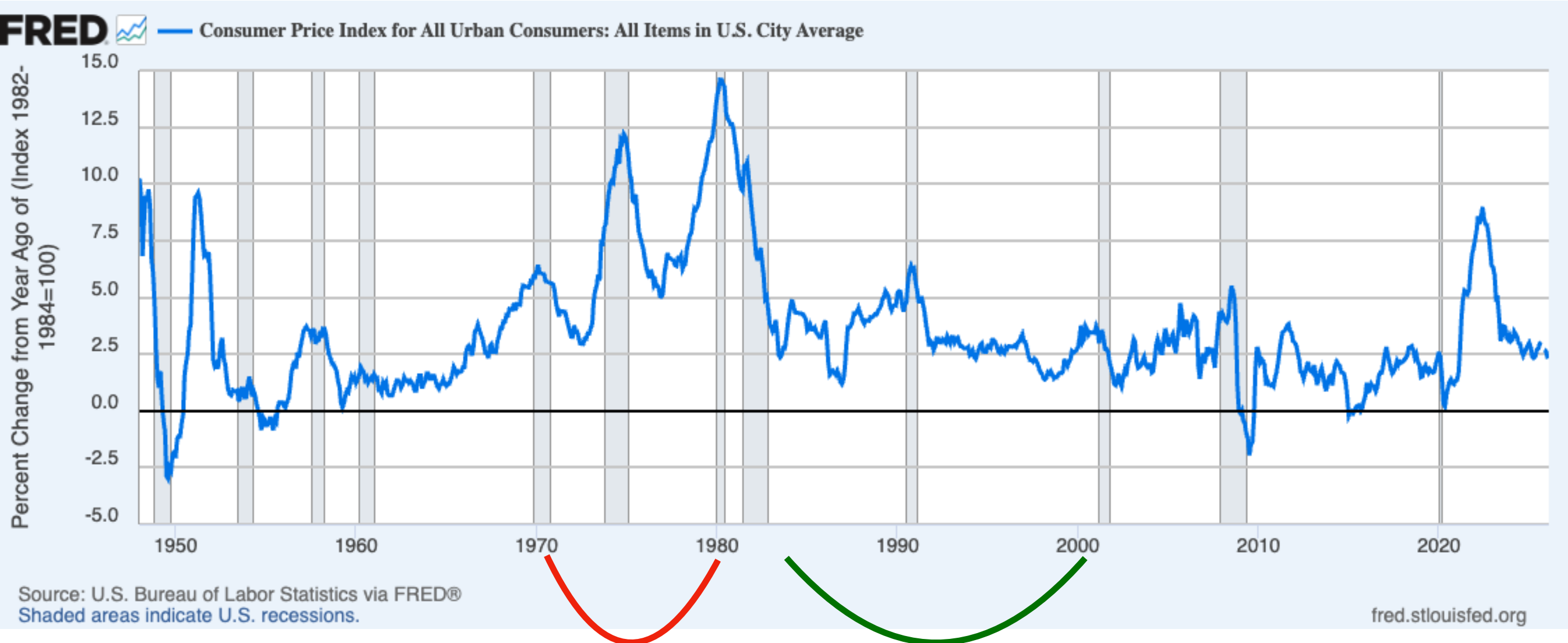
What ARE NOT learning objectives for Chapter 13

- The chapter develops an aggregate demand & aggregate supply model
 - It's cool
 - It's complicated
 - I've felt that its marginal benefit exceeds its marginal cost
- But it develops this algebraically using a **monetary policy rule**, and that is useful to think about

How to think through the short-run model

- We care about
 - **Unemployment** and recessions: $\tilde{Y} < 0$
 - Inflation that is too high: $\pi > 0.02$ with target $\bar{\pi} = 0.02$
- Demand or supply considerations can affect either or both of these
- Exam or essay questions in Econ 100B are going to start with something about changes in demand or supply
- Then your task is to choose the right graph — either the IS-MP diagram or the Phillips curve PC diagram — and start modeling
 - What happens first
 - How the Fed reacts with monetary policy to try and obtain and maintain $\pi = \bar{\pi} = 0.02$

Using IS-MP-PC to understand macroeconomic history of the 1970s and 1980s

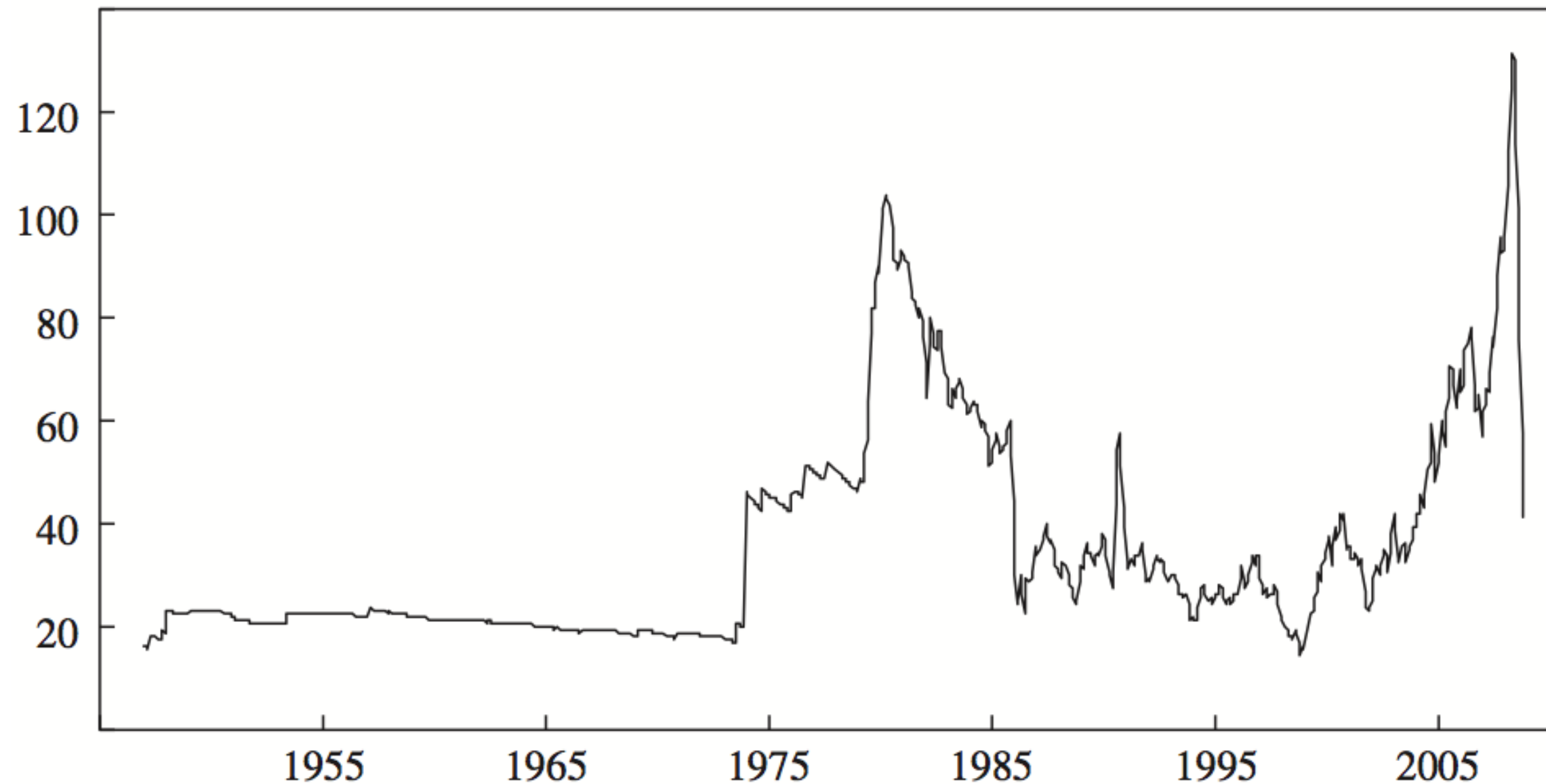


- Why did high inflation become such a **problem** in the 1970s?
- How did policymakers **regain control** over inflation and bring it back down to levels not seen since the 1960s?

Per [Hamilton \(BPEA, 2009\)](#), 1970s oil shocks were big

Figure 1. Real Crude Oil Prices, January 1947 to December 2008^a

Constant dollars per barrel

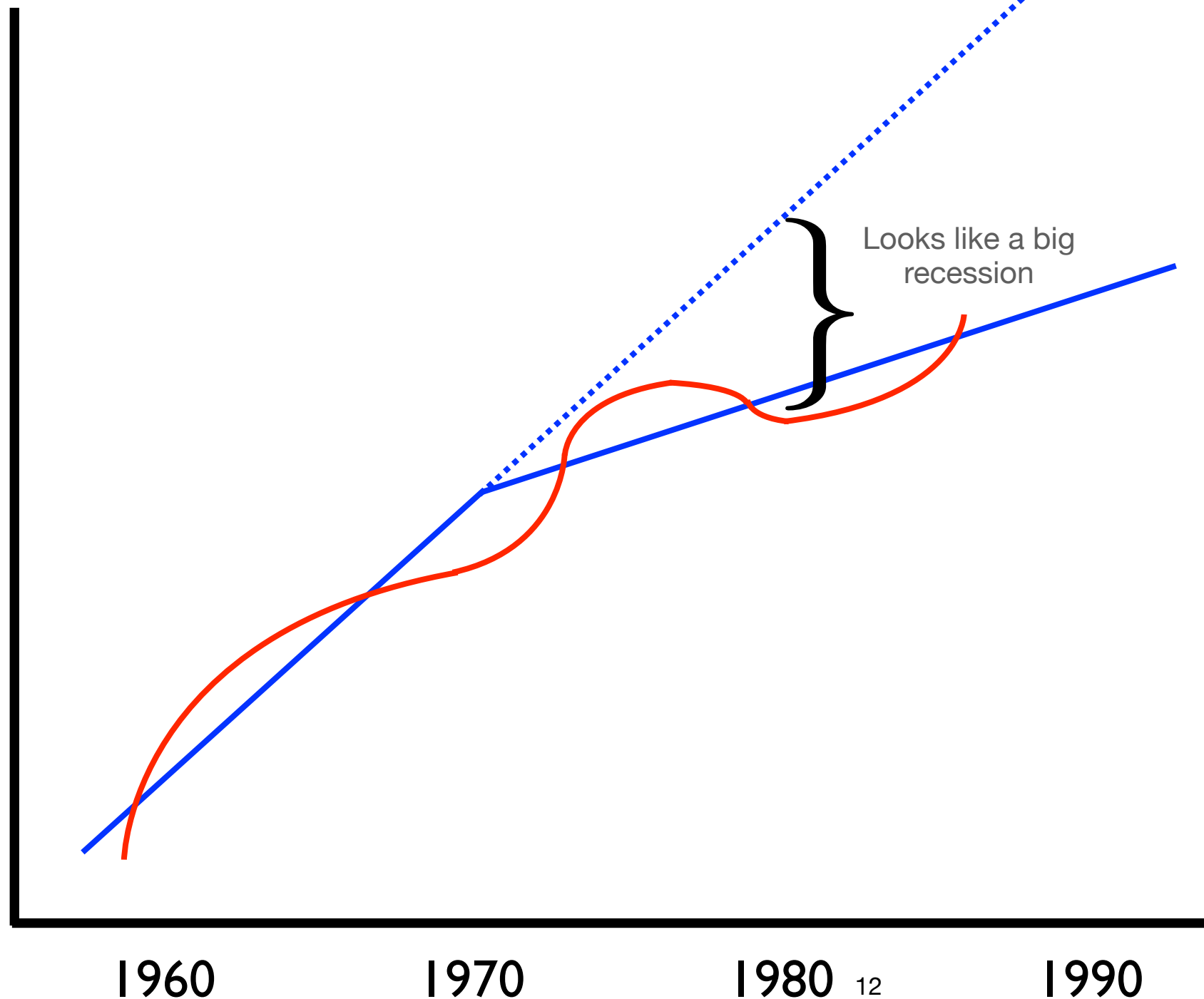


Sources: Energy Information Administration (EIA) data; Bureau of Labor Statistics data.

a. Monthly average price of West Texas Intermediate crude in dollars of November 2008.

And growth in TFP

Real output,
ratio scale



- Recall that

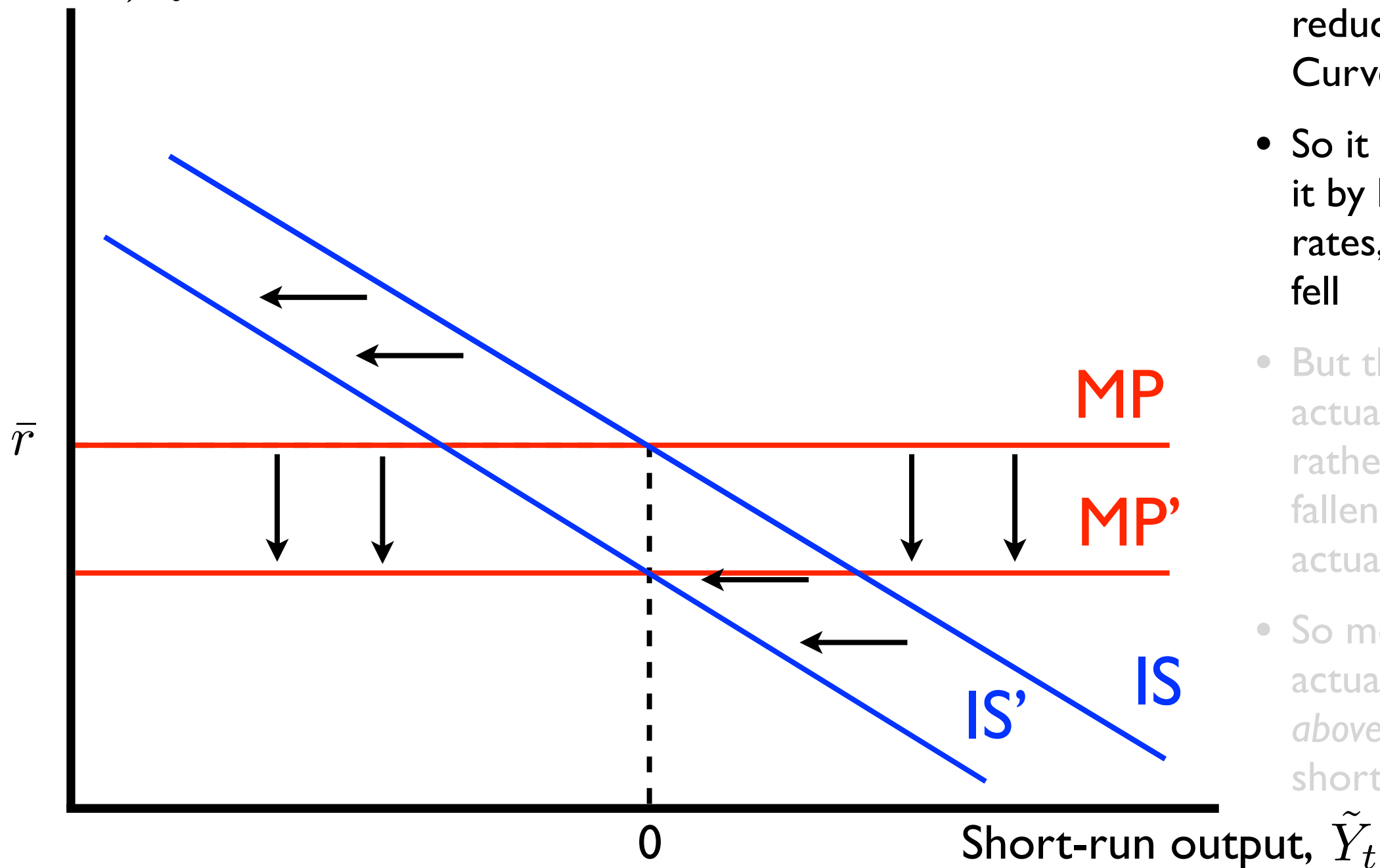
$$\bar{Y} = AK^{1/3}L^{2/3}$$

$$g[Y] = g[A] + \frac{1}{3}g[K] + \frac{2}{3}g[L]$$

- Starting around 1970 or 1973, $g[A]$ decreased
- But policymakers misunderstood this as a big recession

Seeing what looked like a big recession, the Fed “loosened” monetary policy in response

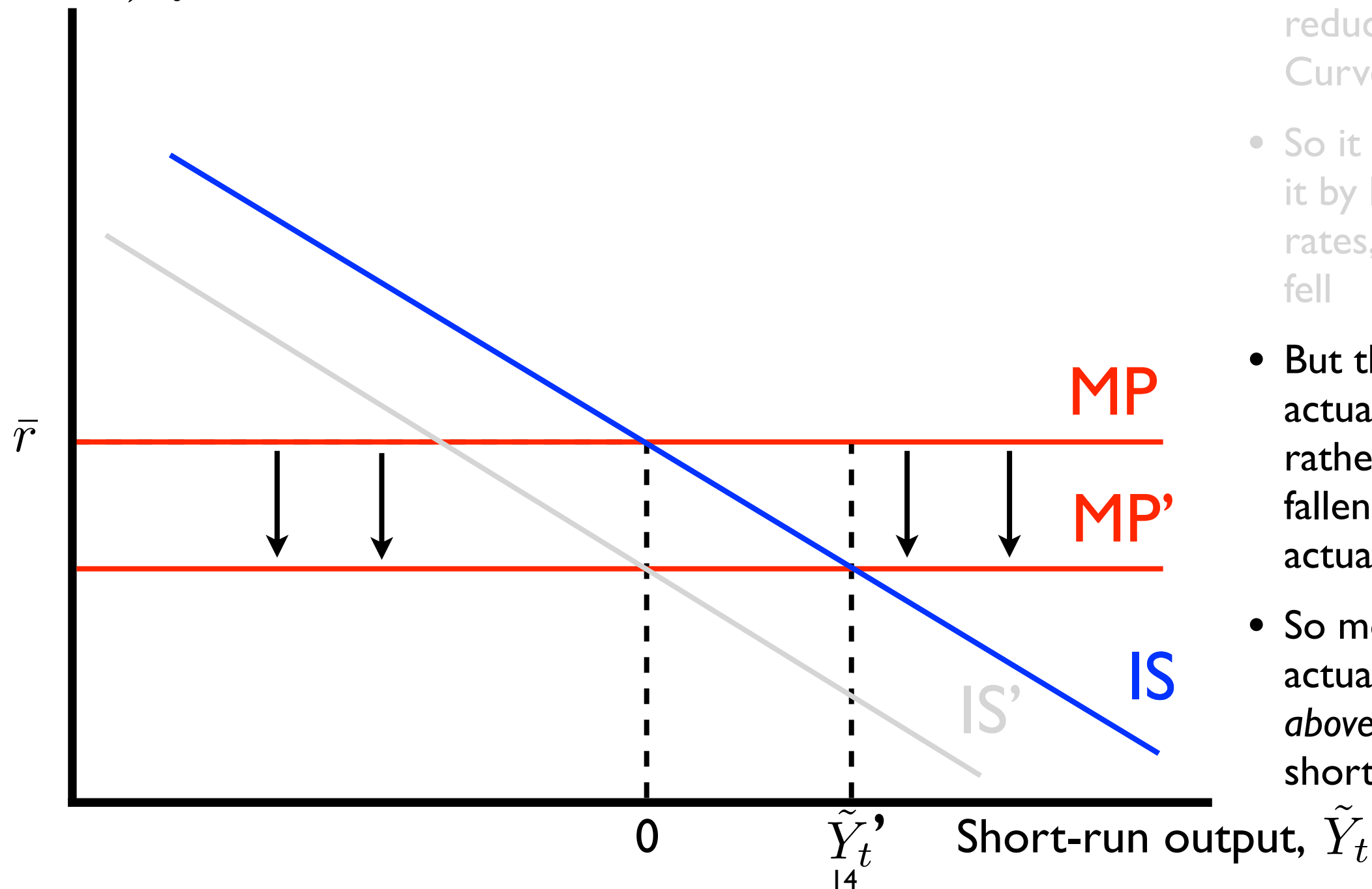
Real interest rate, R_t



- The Fed thought it saw a big recession, a reduction in the IS Curve
- So it tried to counteract it by lowering interest rates, and the MP curve fell
- But the economy wasn't actually in a recession; rather, potential has fallen. The IS Curve actually never moved
- So monetary policy is actually pushing output *above potential* in the short run!

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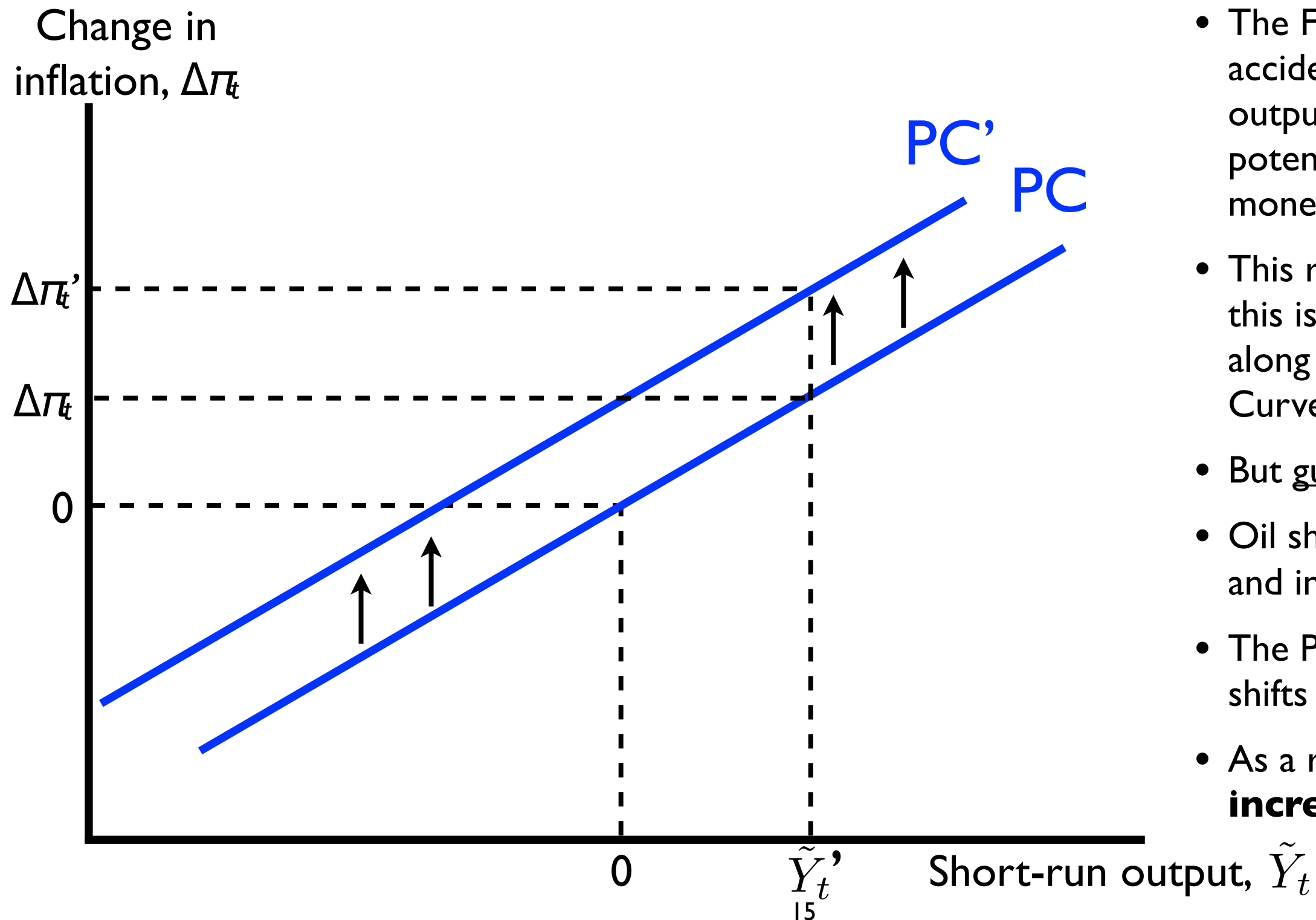
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This would be movement along the Phillips Curve

$$\Delta\pi_t = \bar{v}\tilde{Y}_t + \bar{o}$$



- The Fed has accidentally stimulated output above potential with “easy” money
- This raises inflation; this is movement along the Phillips Curve
- But guess what else?
- Oil shocks also hit and increase $\bar{o}$
- The Phillips Curve shifts upward as well
- As a result, inflation **increases a lot**

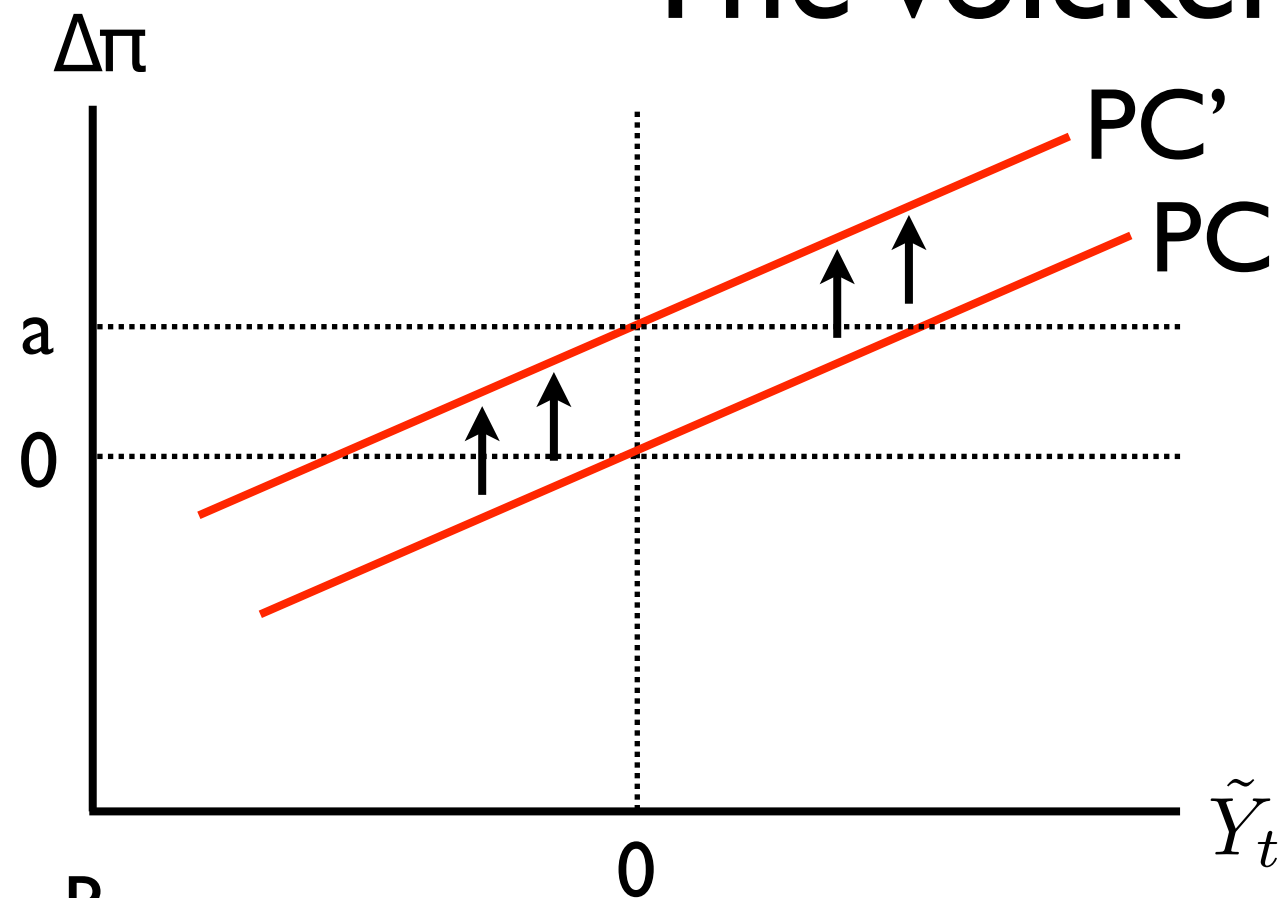
The Volcker Disinflation

- Policymakers agreed that high inflation required action
- [Paul Volcker](#) was nominated by President Jimmy Carter and [confirmed 98-0 by the Senate on August 2, 1979](#). The former chair, G. William Miller, became the Treasury Secretary. [History of chairs here](#)
- The August 1979 CPI release for July 1979 would have showed 11.5% inflation since July 1978
- People expected high inflation, having lived through years of it. Inflation was above 3% after September 1972
- The modern Phillips curve tells us

$$\pi_t = \pi_t^e + \bar{v}\tilde{Y}$$

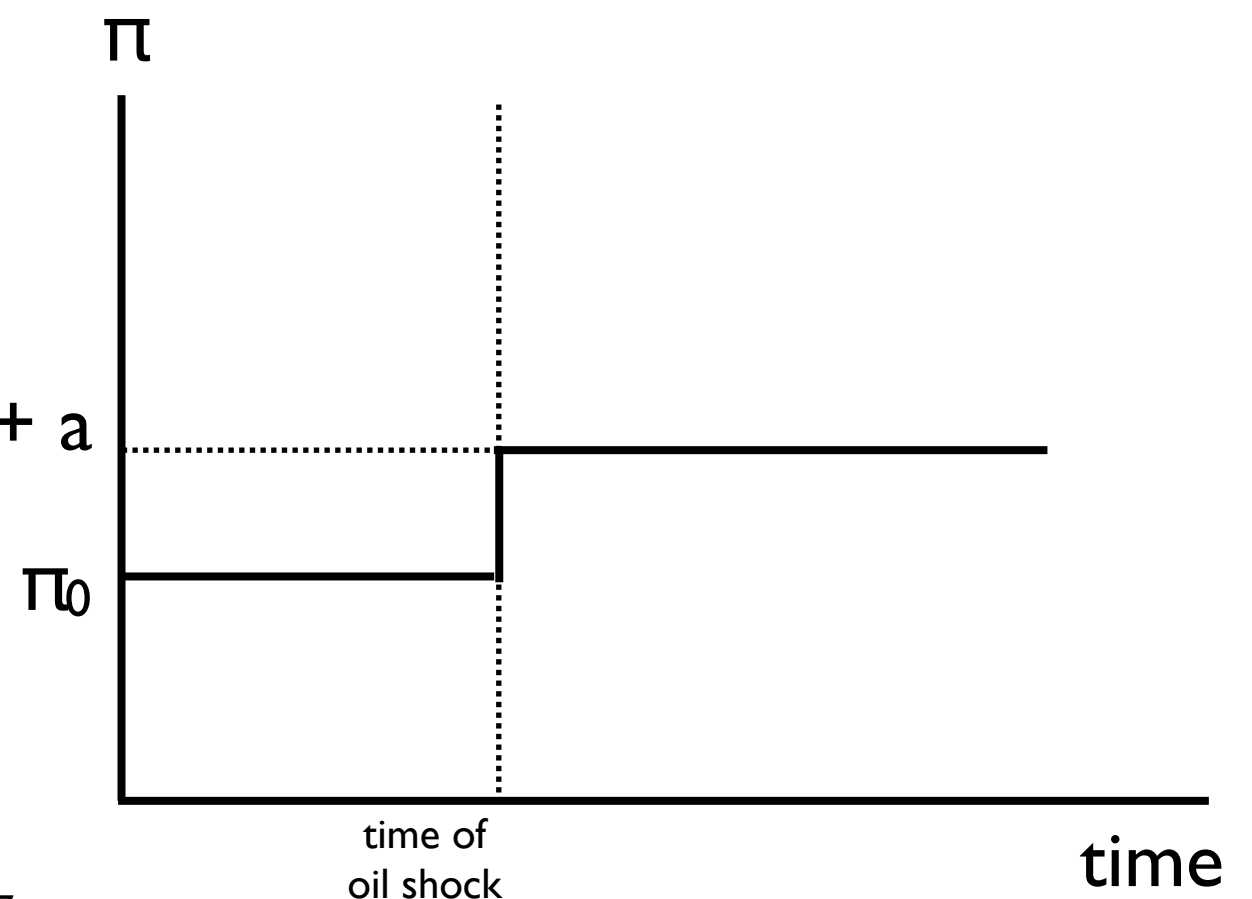
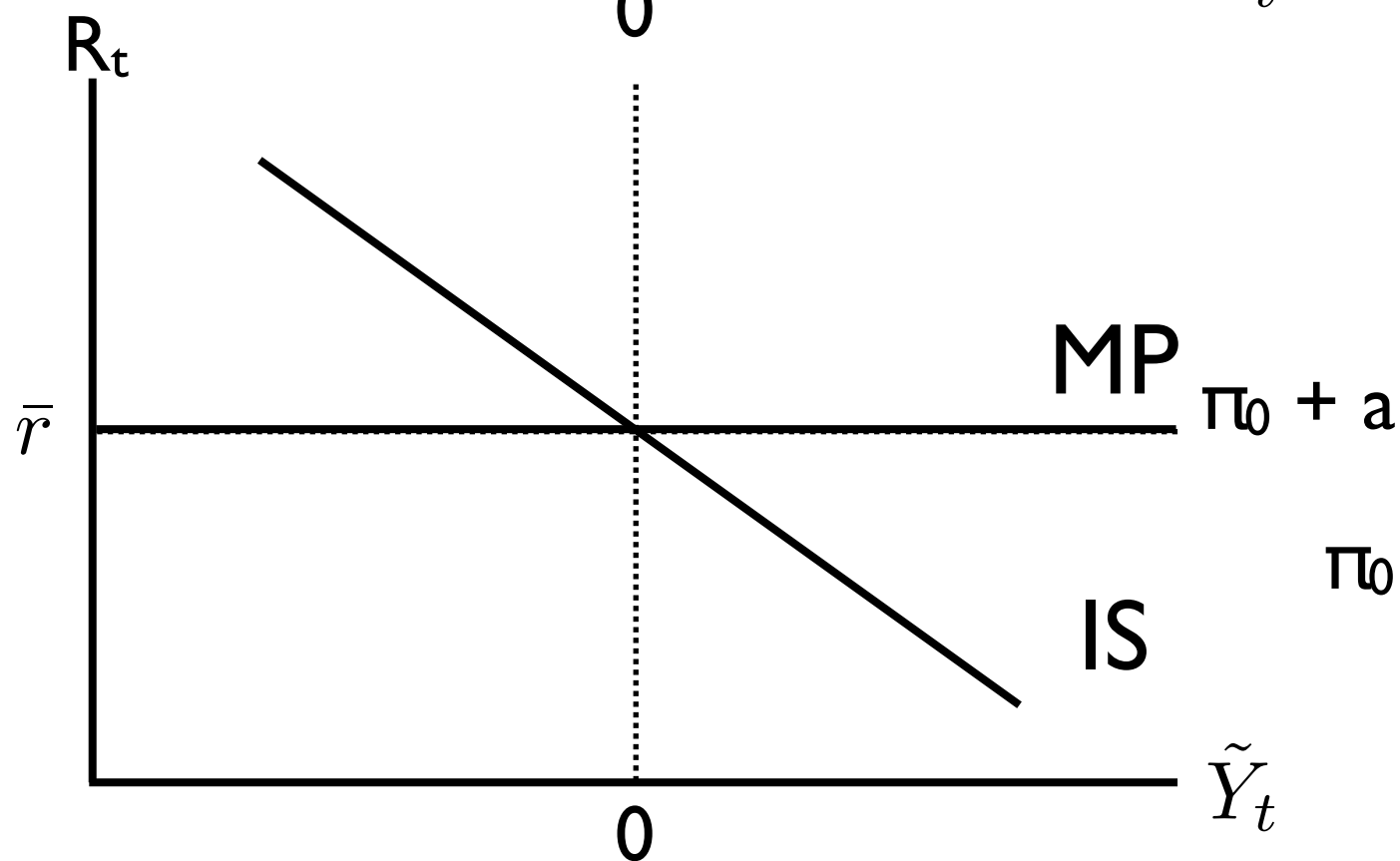
- When π^e is very high, the only thing the Fed can do is raise R high enough to push $\tilde{Y}$ very negative

The Volcker Disinflation

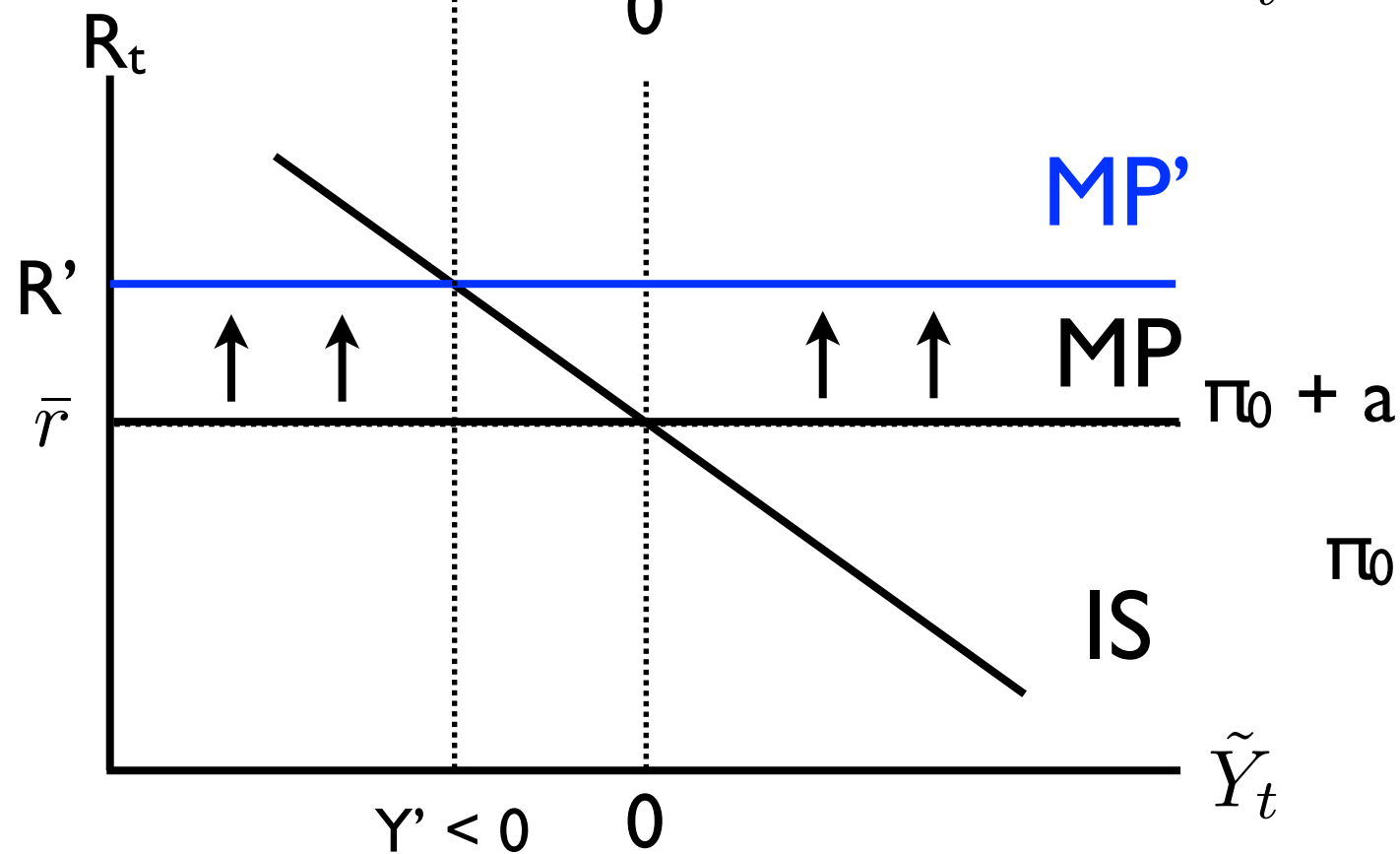
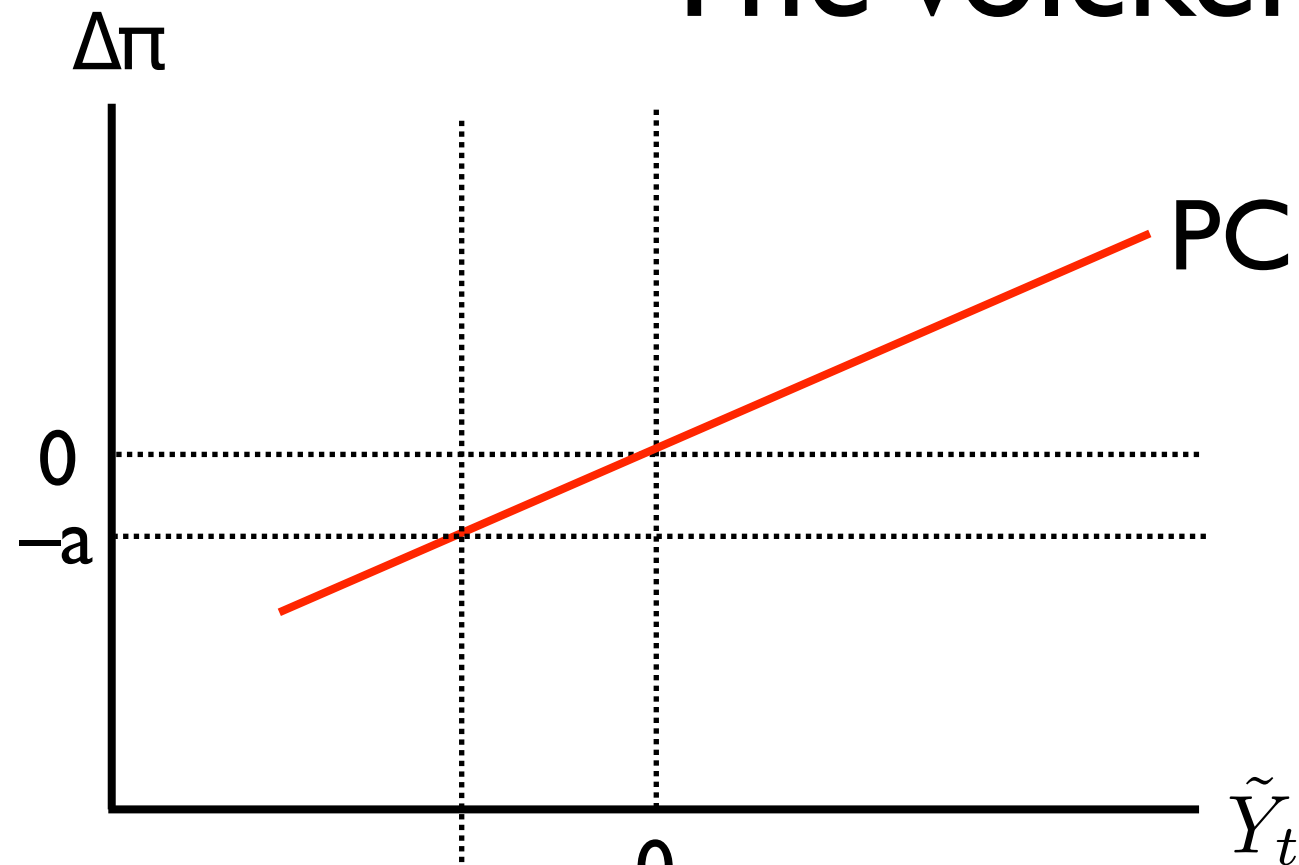


Q: Suppose there's an oil shock that increases all prices and inflation. What happens?

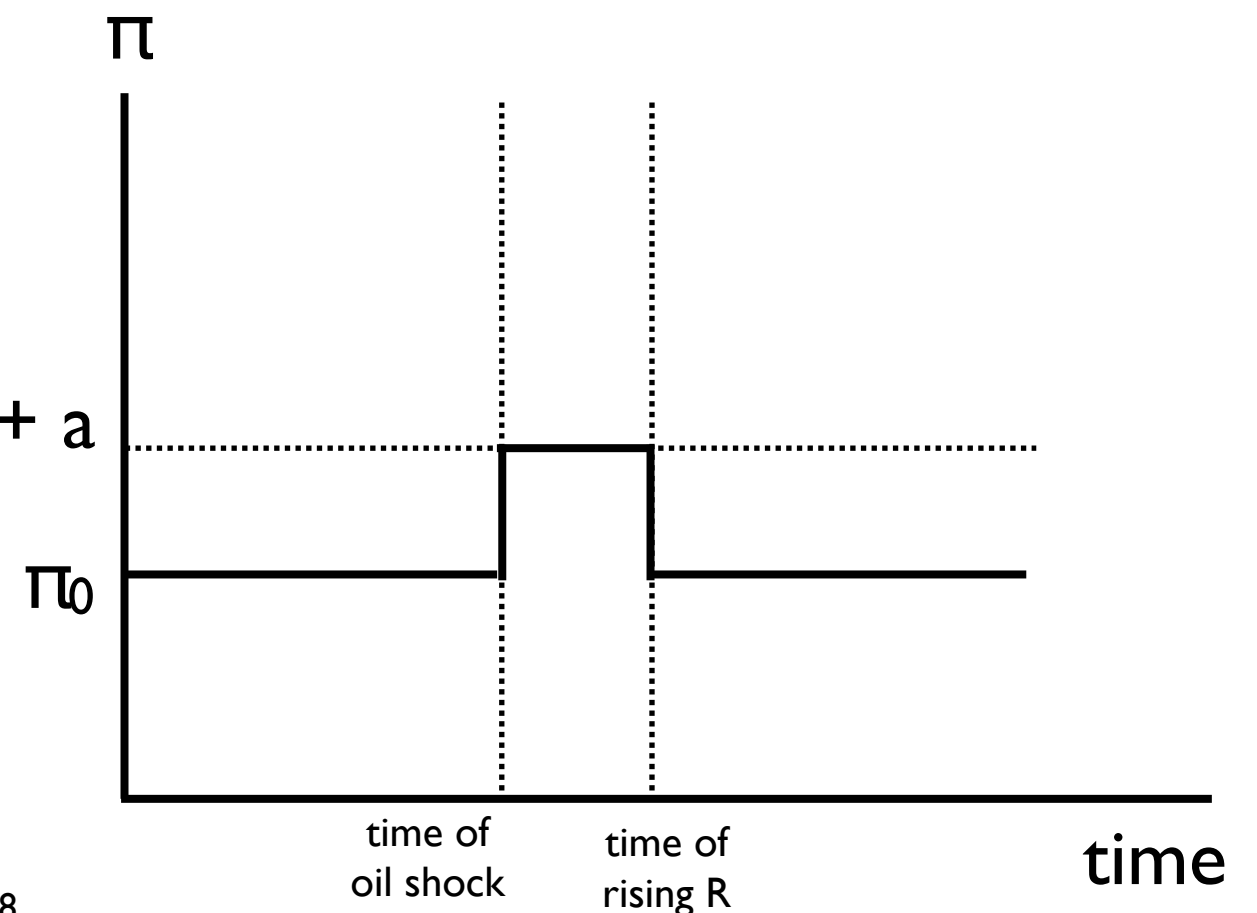
- The Phillips Curve PC jumps up to PC'
- Inflation increases to $\pi_0 + a$
- If nothing else happens, PC' will recede to PC
- And $\Delta\pi = 0$ again
- But the change in inflation would be permanent!



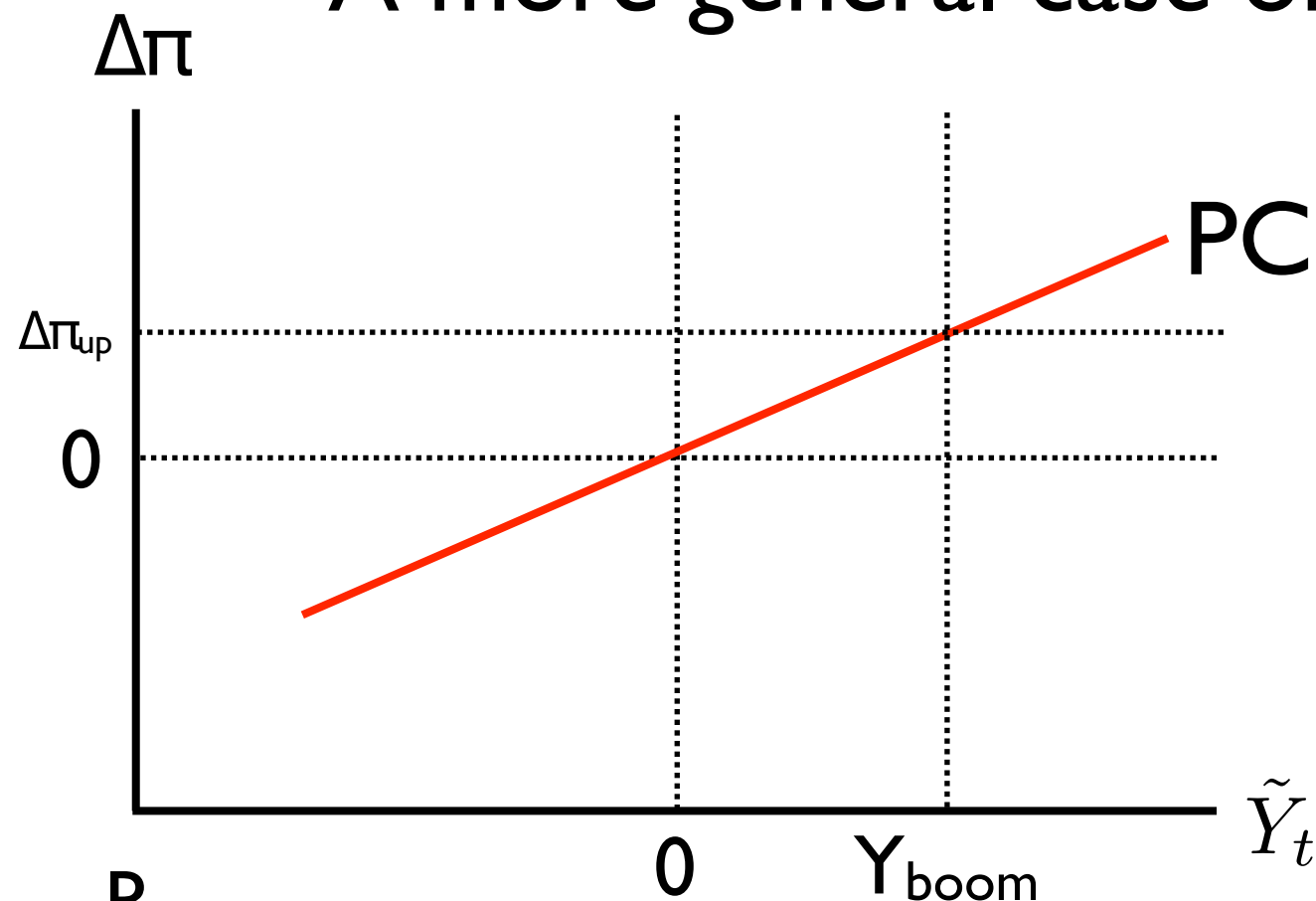
The Volcker Disinflation



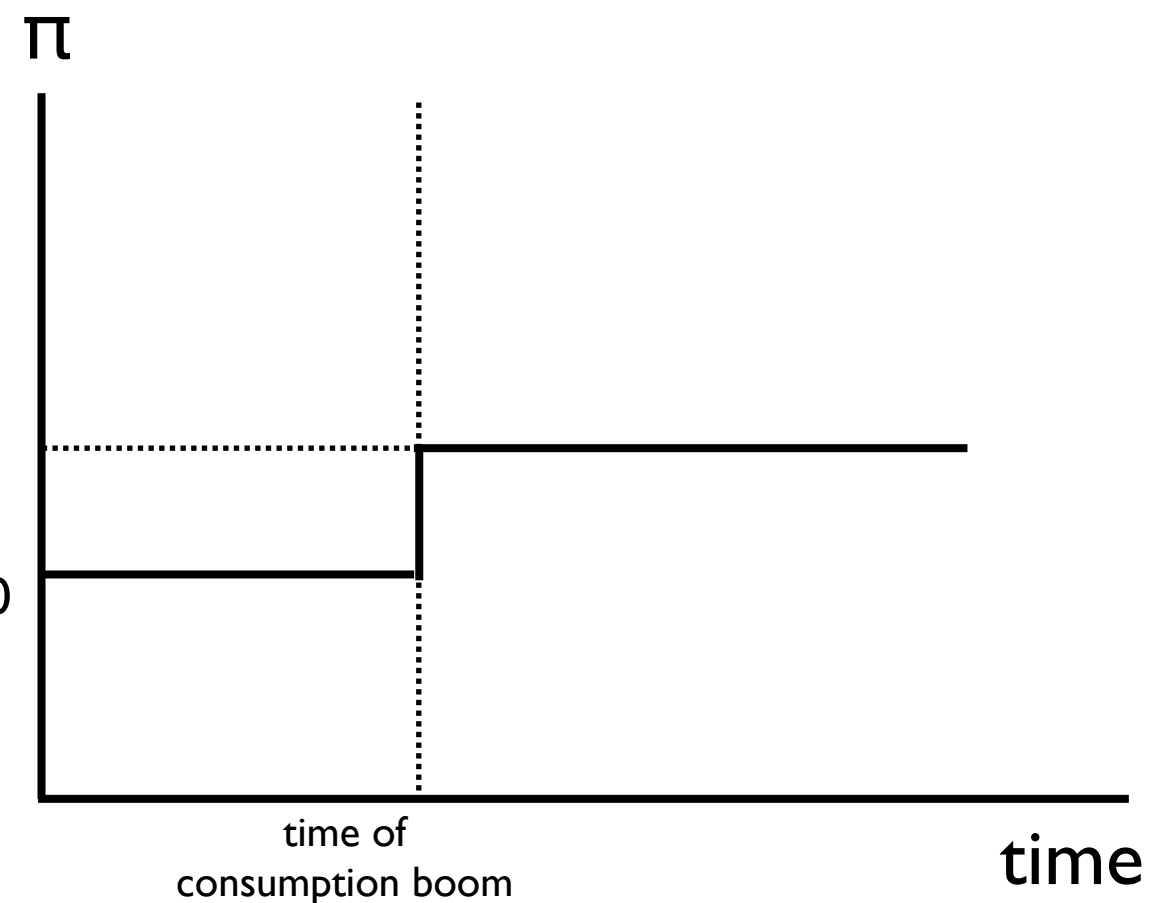
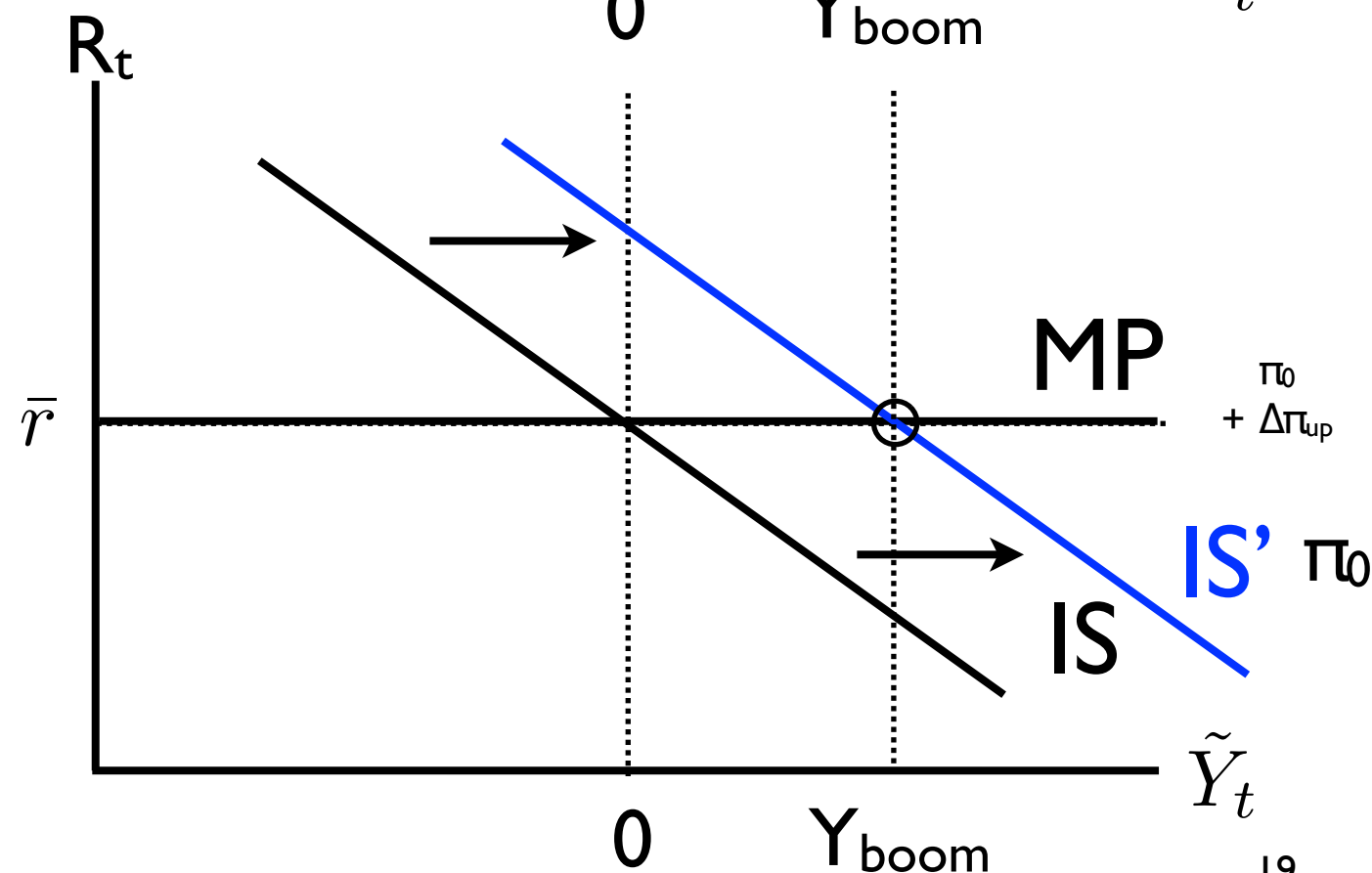
- Inflation rose because the Phillips Curve shifted
- The Fed must act to reduce inflation in the only way that it can: by causing a recession
- The Fed raises the real interest rate R by raising the MP Curve, causing a recession
- The recession then lowers inflation through the Phillips Curve
- Once inflation has fallen, the Fed lowers the real interest rate, and the recession ends



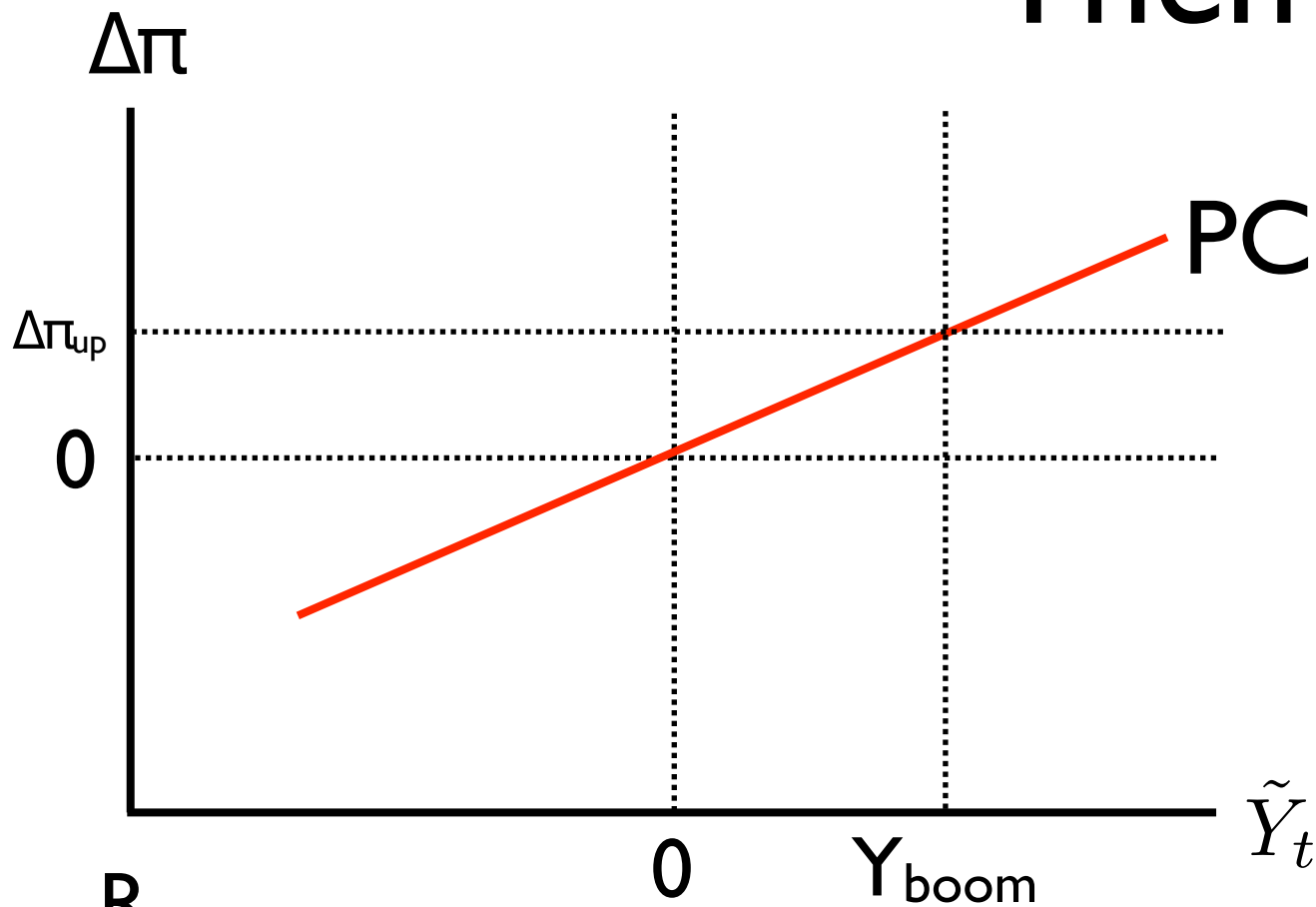
A more general case of “demand management”



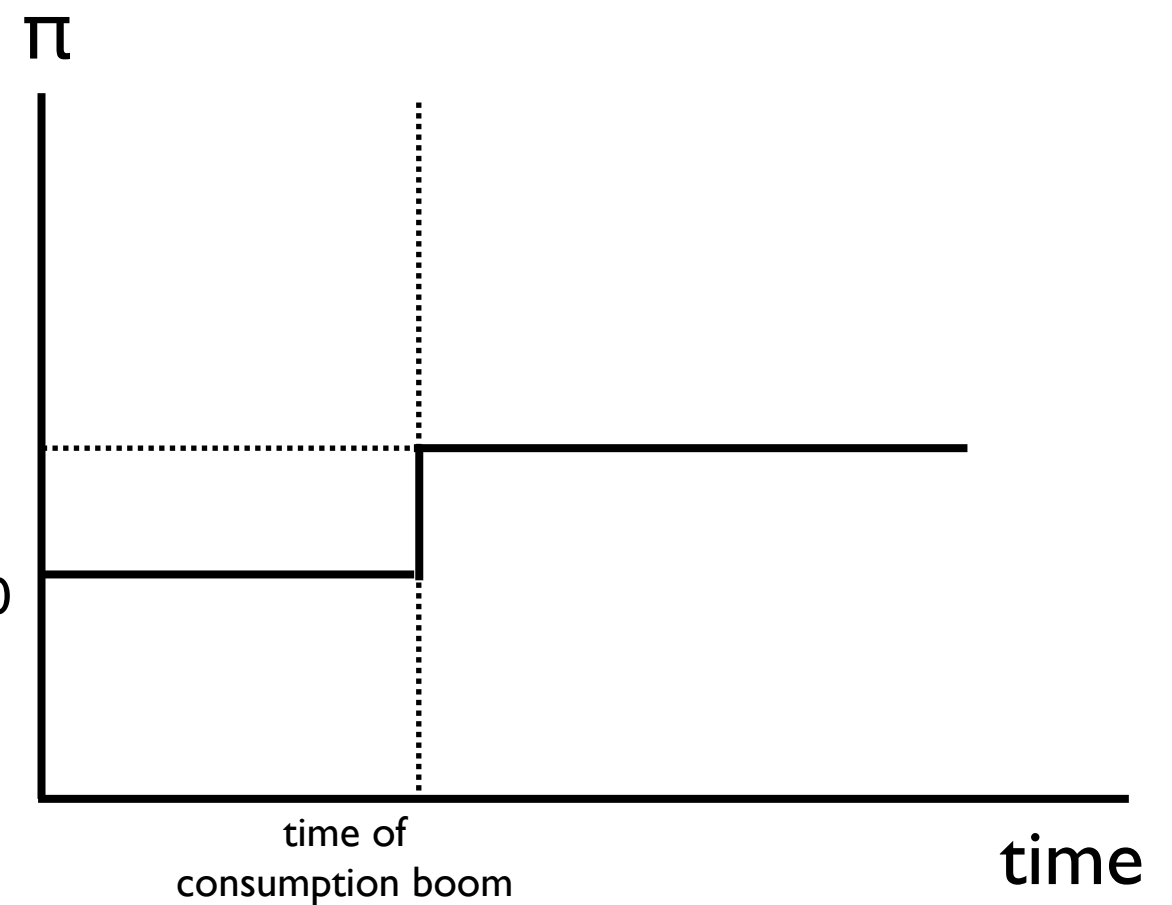
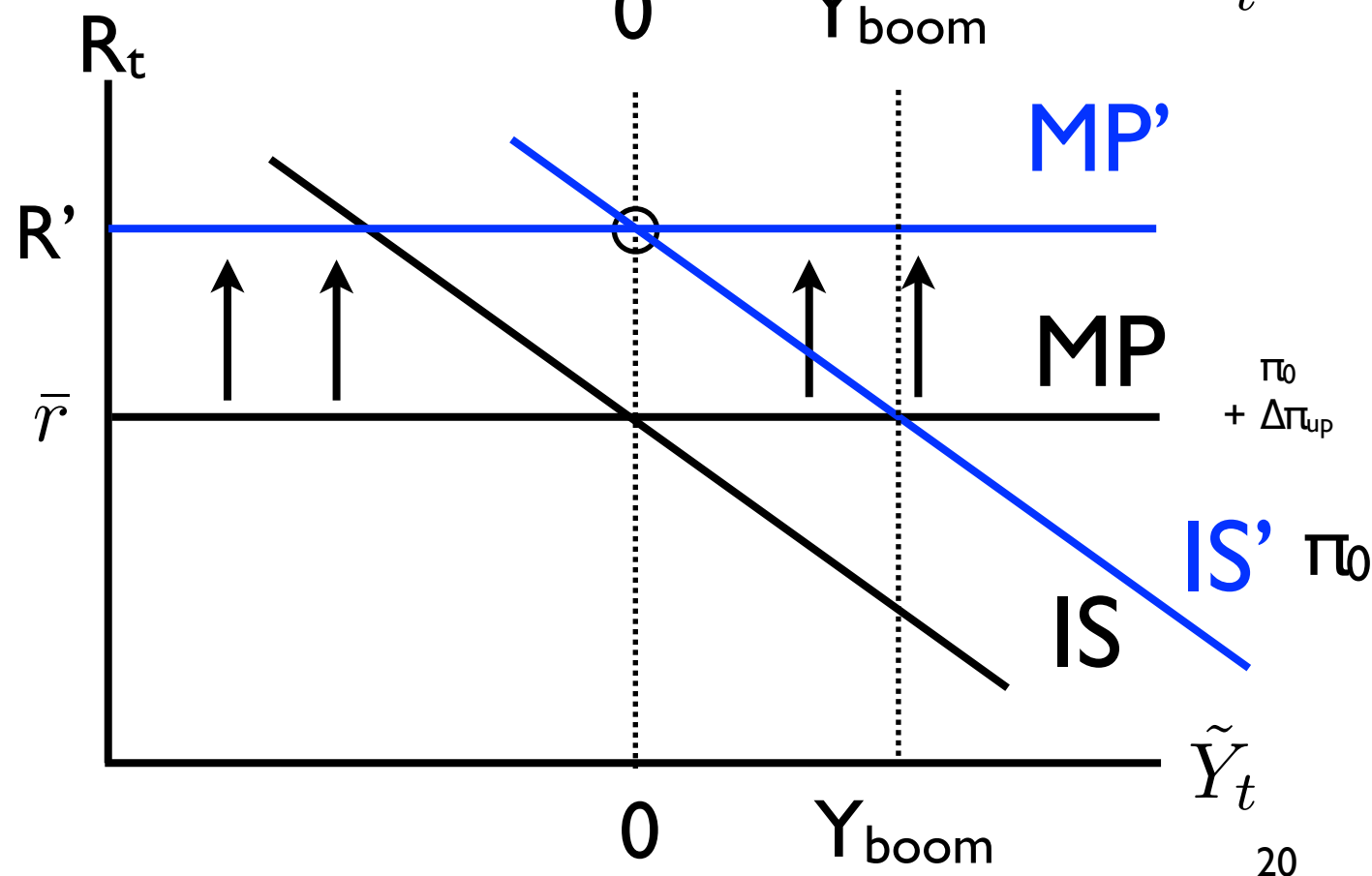
- There are many things that could shock the economy either negatively or positively
- Suppose U.S. consumers decide they want to consume a larger fraction of their permanent incomes. What happens in IS-MP / PC ?
- The IS Curve shifts out, because $\bar{a}_c = C / \bar{Y}$ rises, and the economy booms
- Through the Phillips Curve, inflation rises



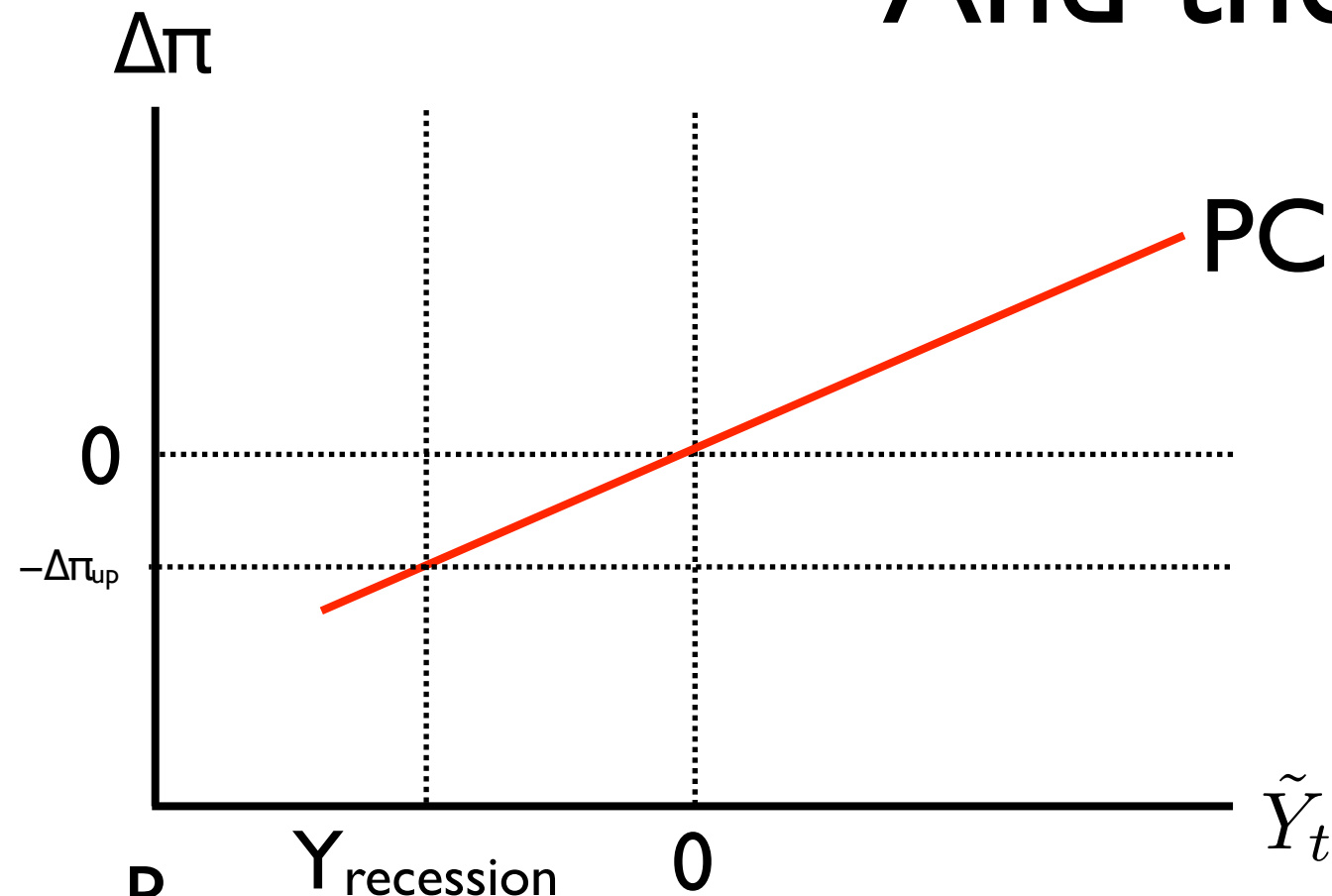
Then what?



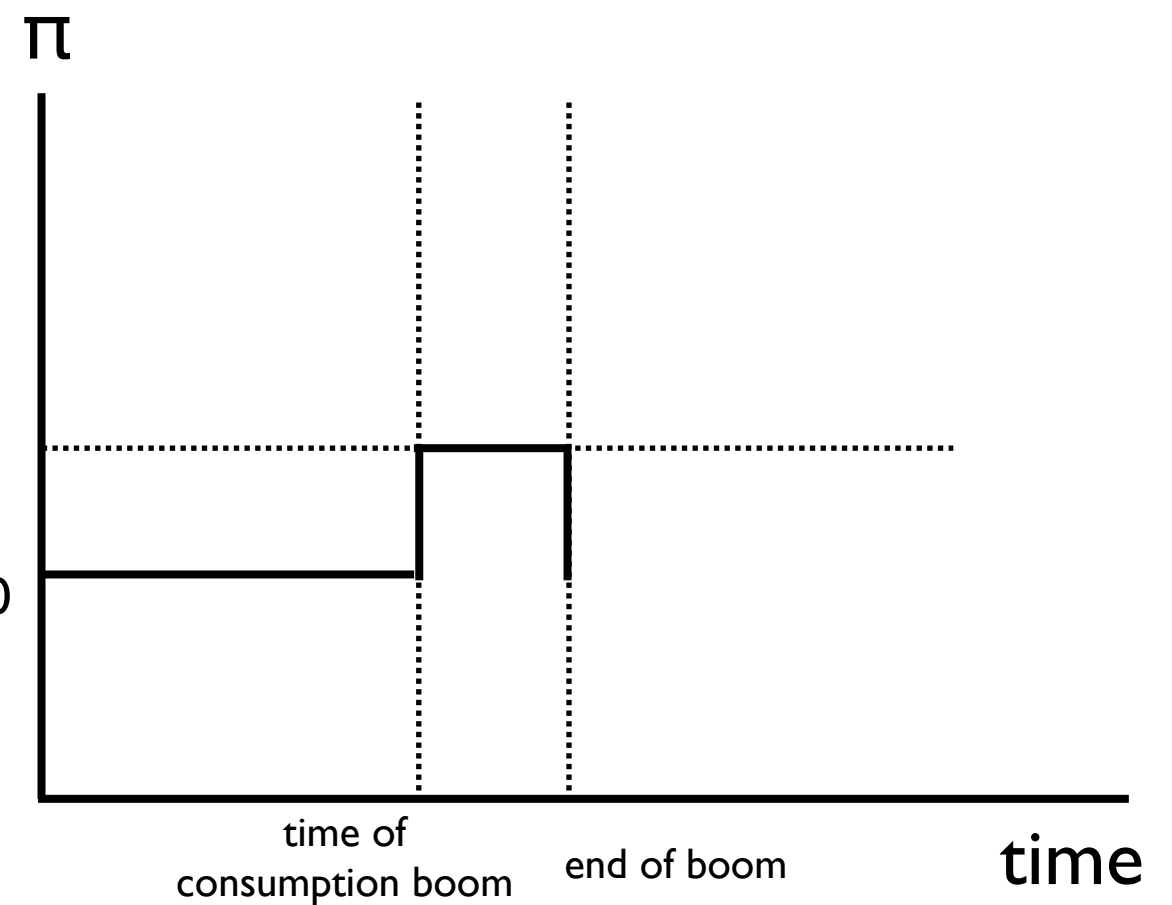
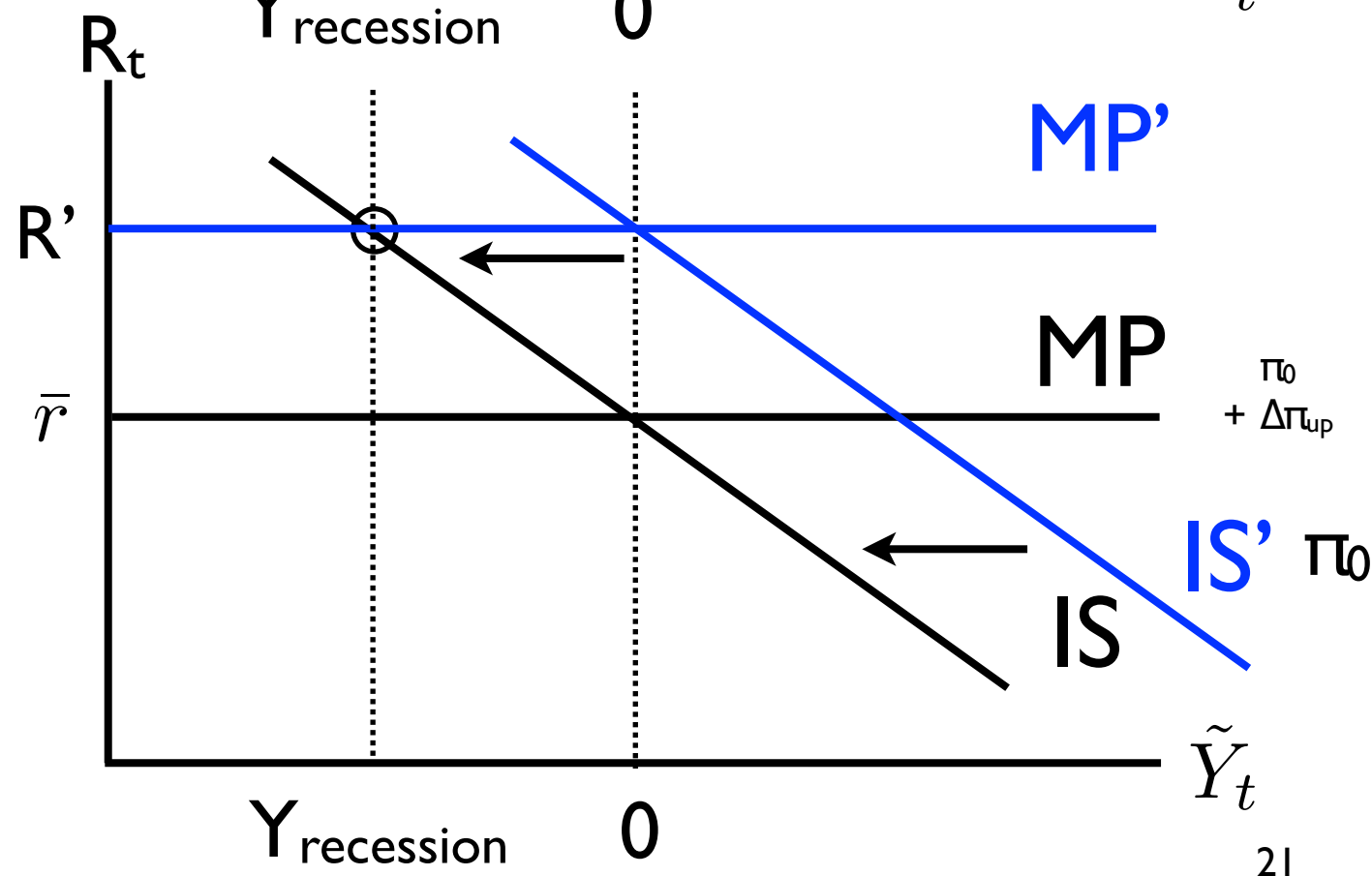
- In response to the rise in inflation above π_0 , the U.S. Federal Reserve will raise interest rates to combat inflation
- The MP Curve shifts upward
- If the increase is large enough, this will stop the rise in inflation



And then what?



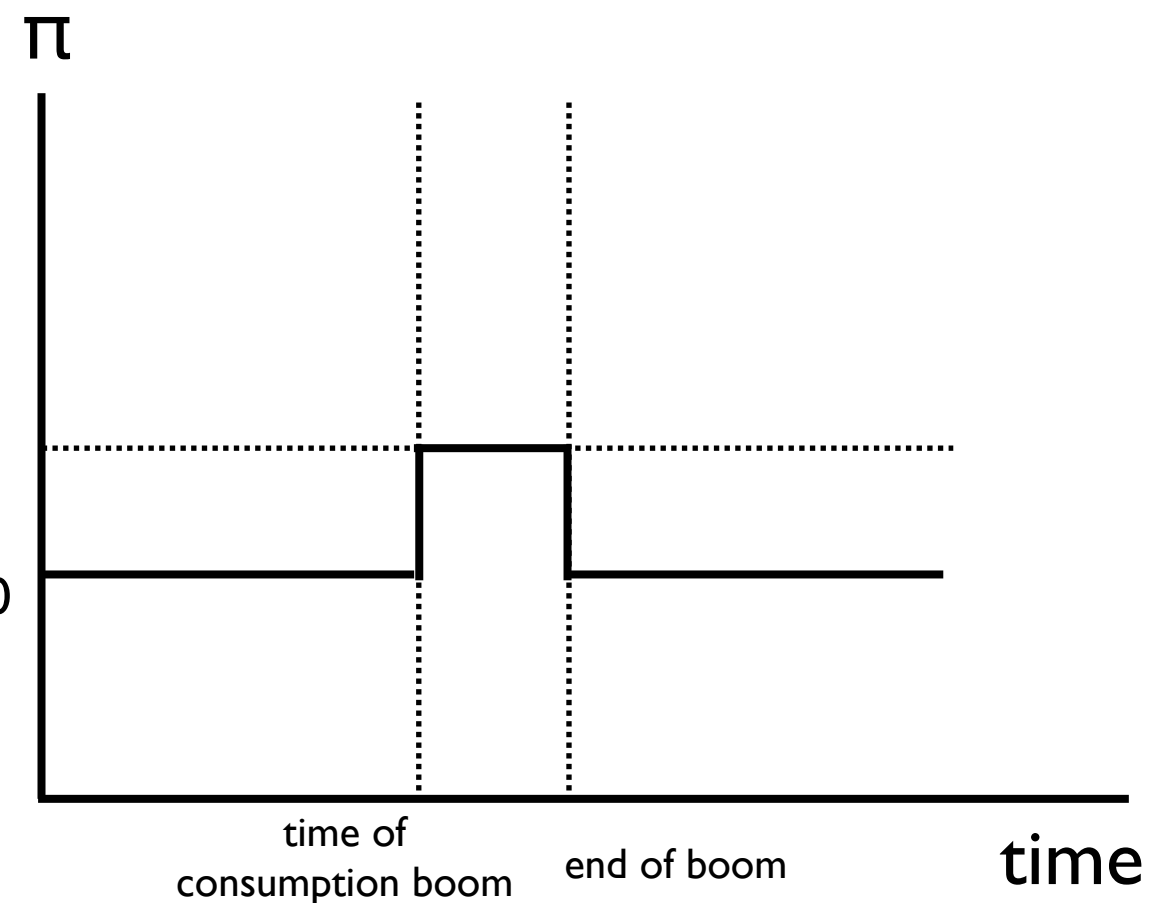
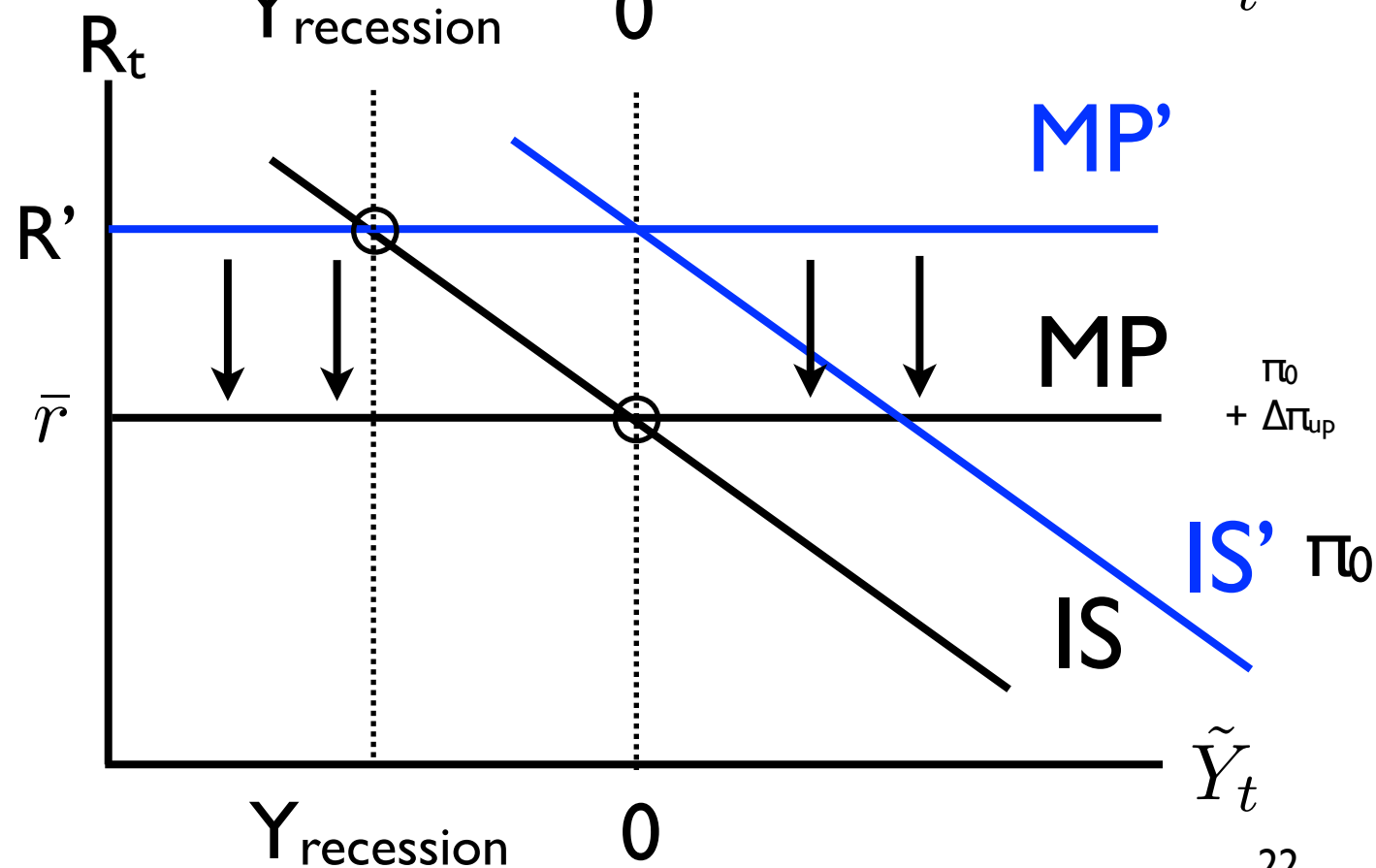
- Eventually, ***all*** positive shocks to aggregate demand must wear off; Consumption must return to its normal share of permanent income
- The IS Curve shifts back leftward, causing a recession because the interest rate is high
- This *lowers inflation* through the Phillips Curve



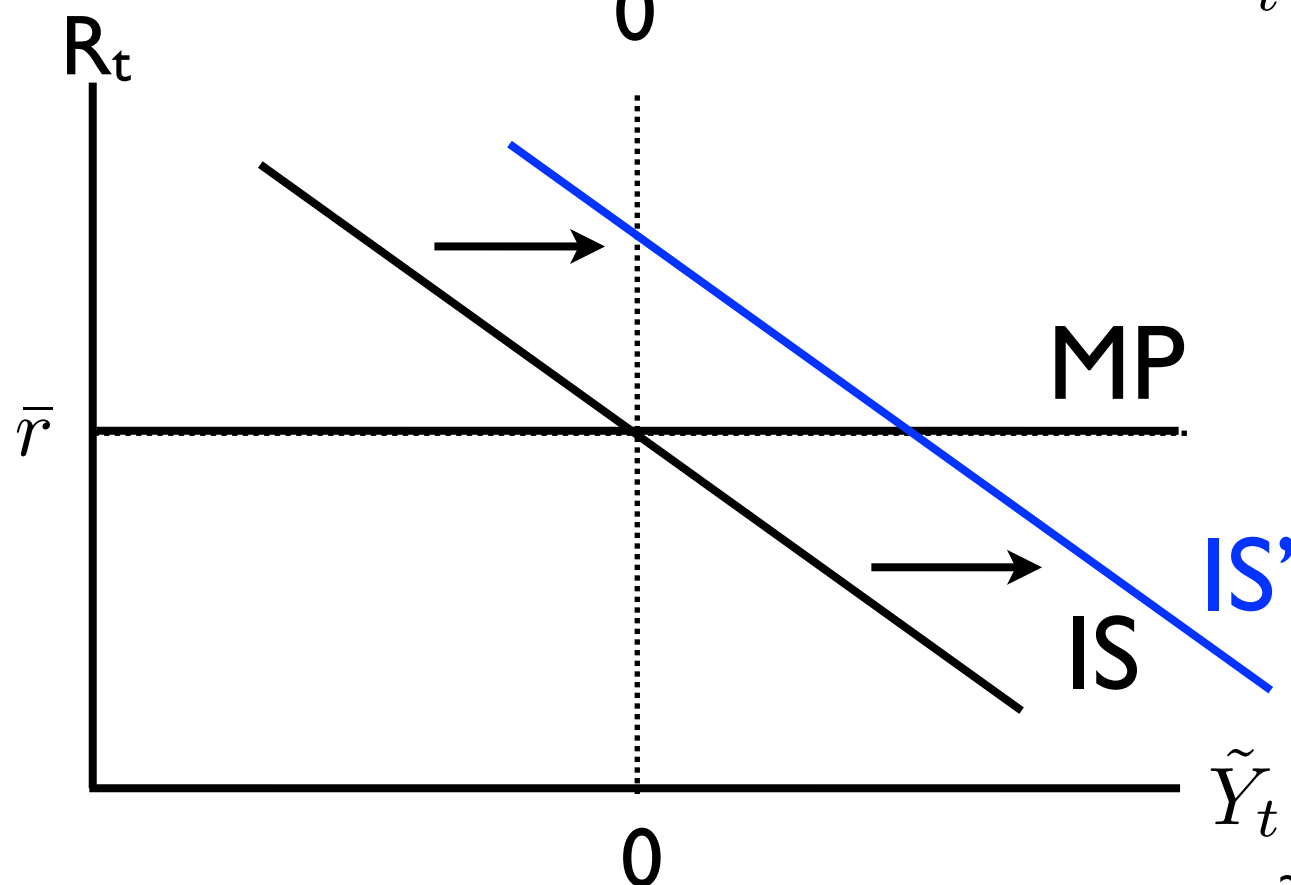
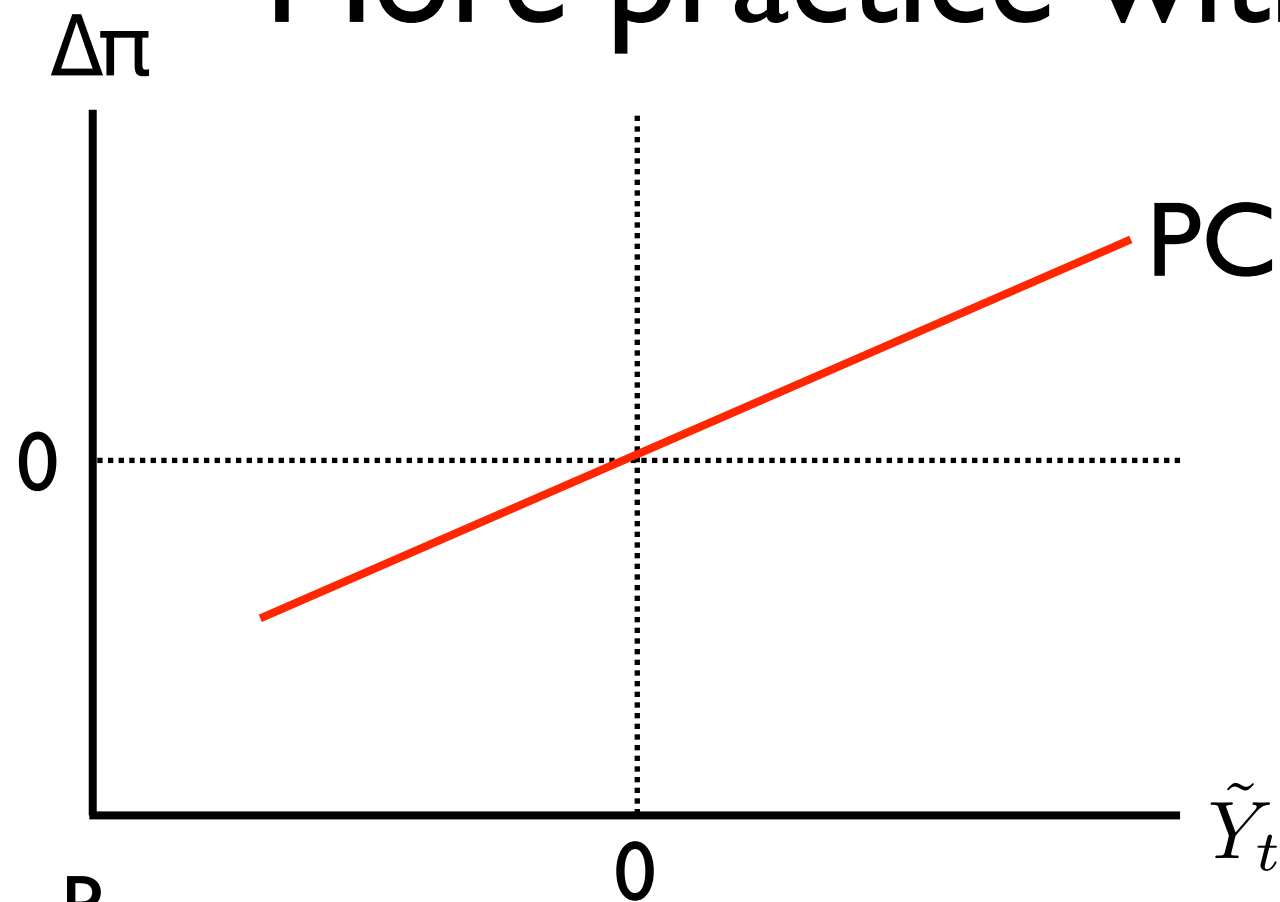
Then the Fed can lower interest rates



- Once the shock has worn off and inflation has returned to its normal level, the Fed lowers interest rates back to normal
- The MP Curve falls, and we're back where we started, at short-run output equal to zero, neither a boom nor a bust
- Through the Phillips Curve, inflation is stable



More practice with IS-MP & PC (1/3)

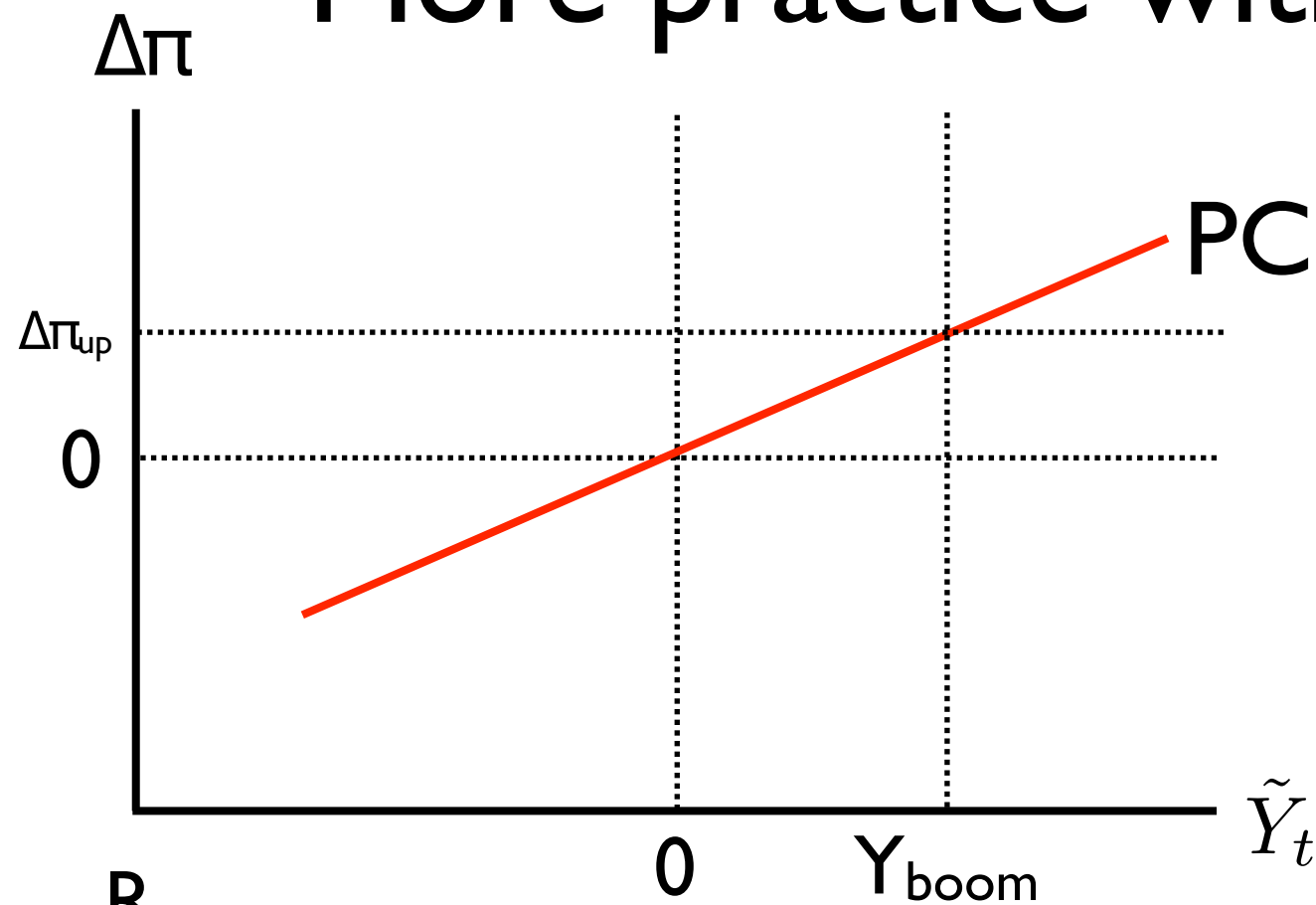


Q: Suppose the government expands its spending temporarily in order to stimulate the economy. What happens in the IS-MP or PC diagrams?

$$\tilde{Y}_t = \bar{a} - \bar{b} (R_t - \bar{r})$$

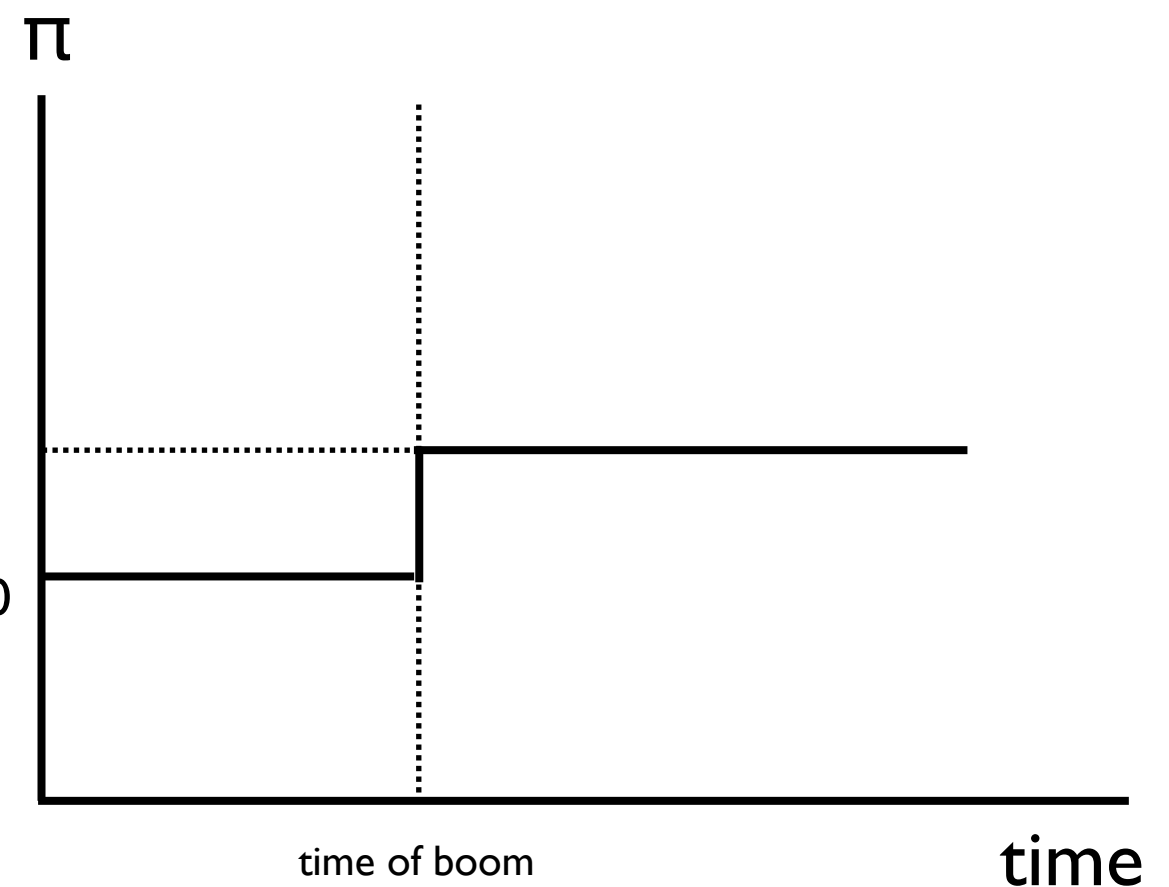
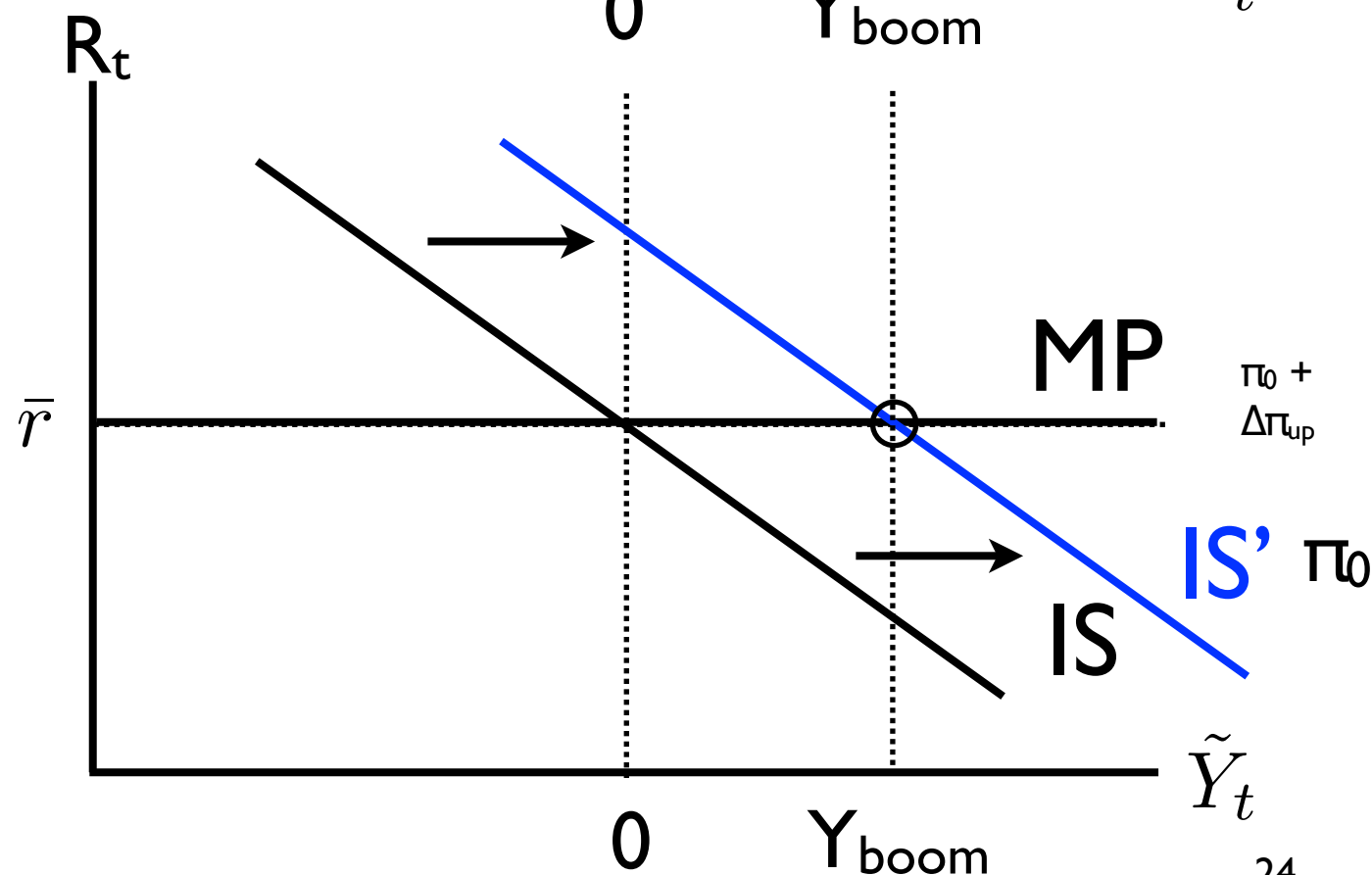
- The government temporarily spends more than its usual share of permanent income $\bar{a}_G = 0.20$
- When $\bar{a}_G$ rises, that pulls $\bar{a}$, the intercept of the IS curve, up with it.
- This is a *positive* aggregate demand shock of an amount equal to the increase in $\bar{a}_G$
- It's a **rightward shift** in the IS curve to IS'

More practice with IS-MP & PC (2/3)

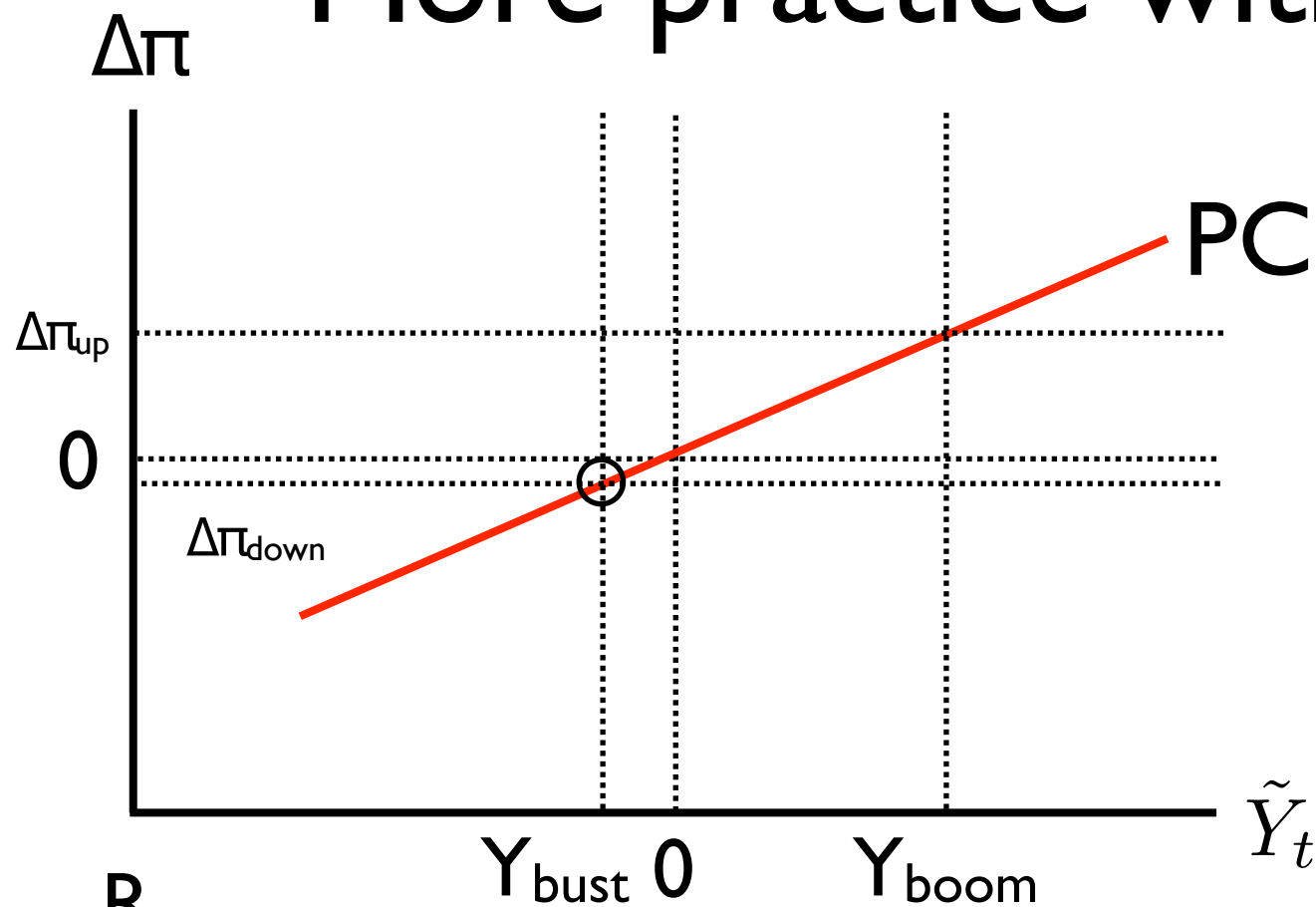


Q: Suppose the government expands its spending temporarily in order to stimulate the economy. What happens in the IS-MP or PC diagrams?

- The IS Curve has shifted to the right
- This produces a boom!
- This means that unless the Fed raises the real interest rate R , inflation will increase!

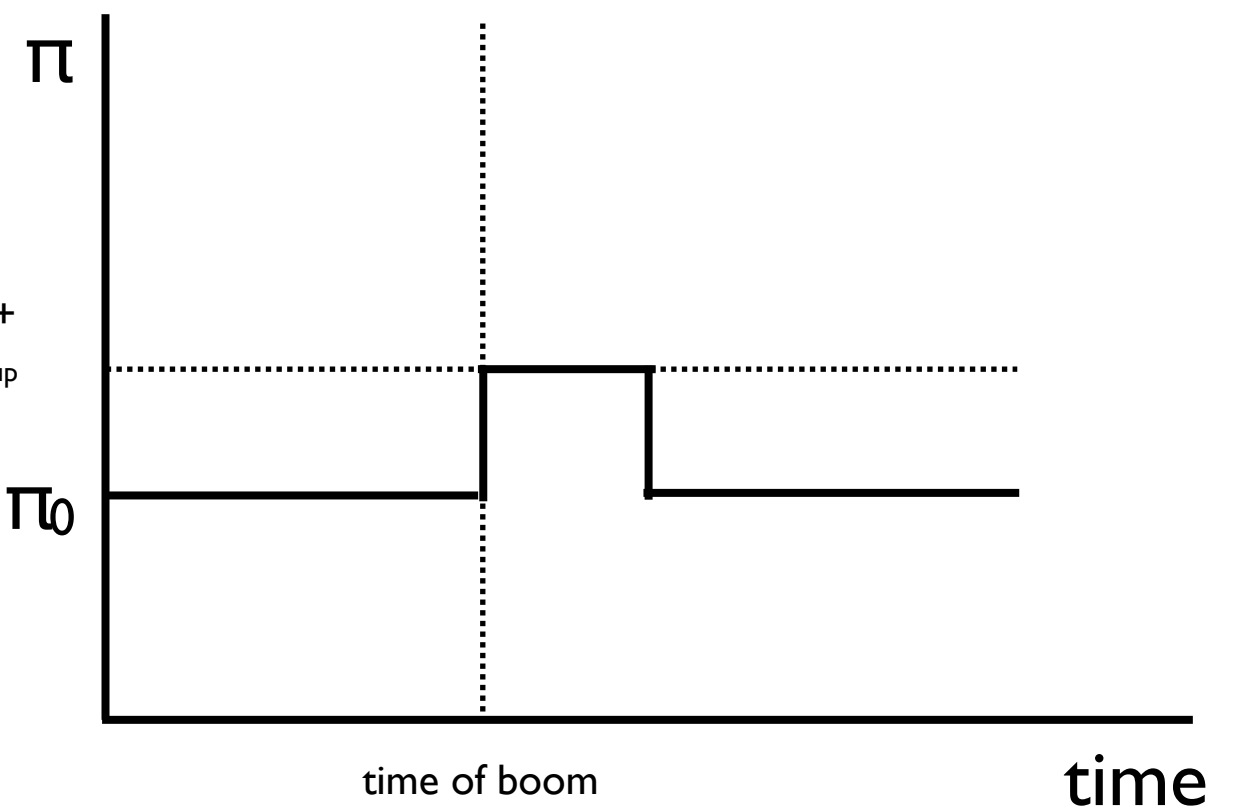
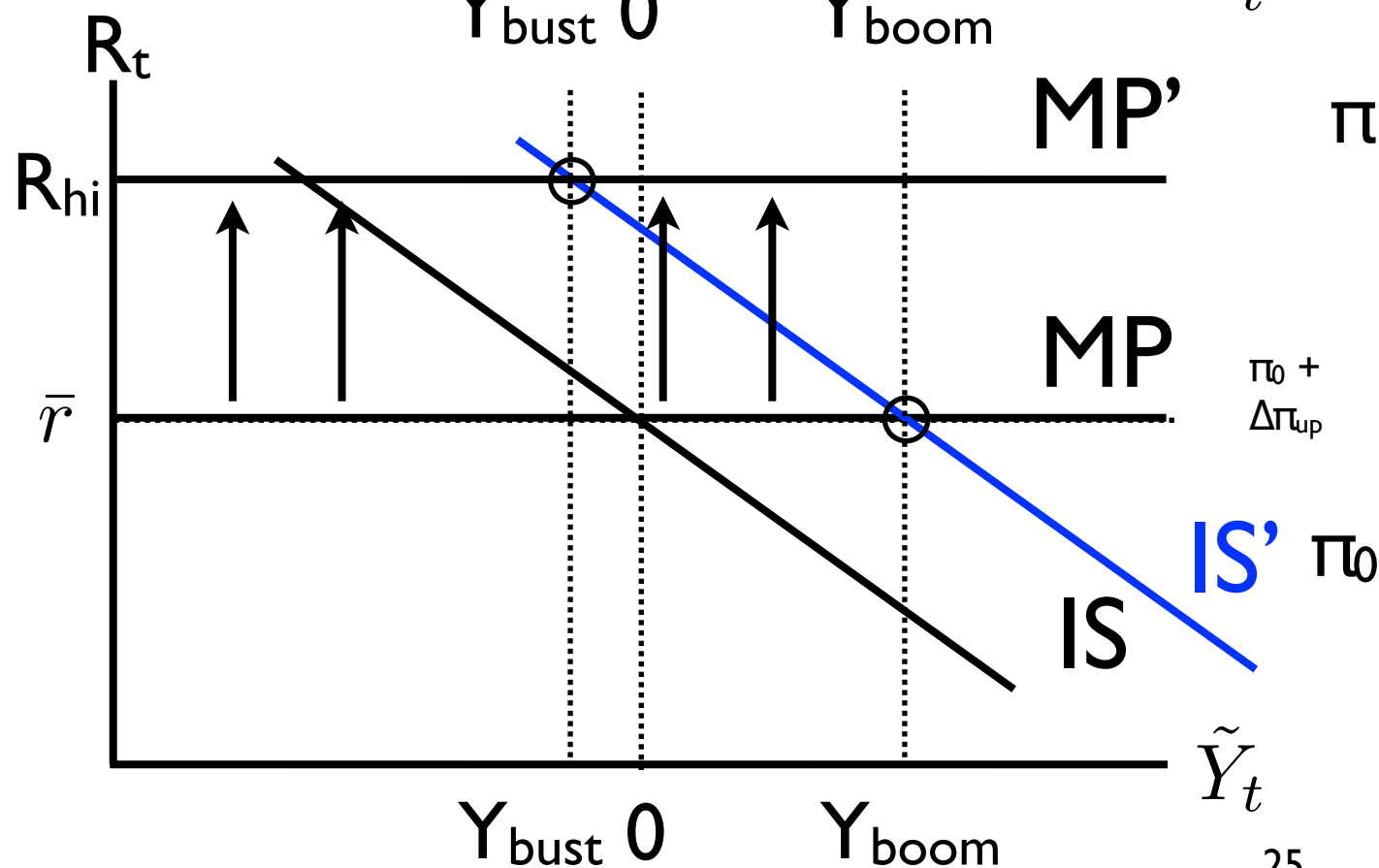


More practice with IS-MP & PC (3/3)

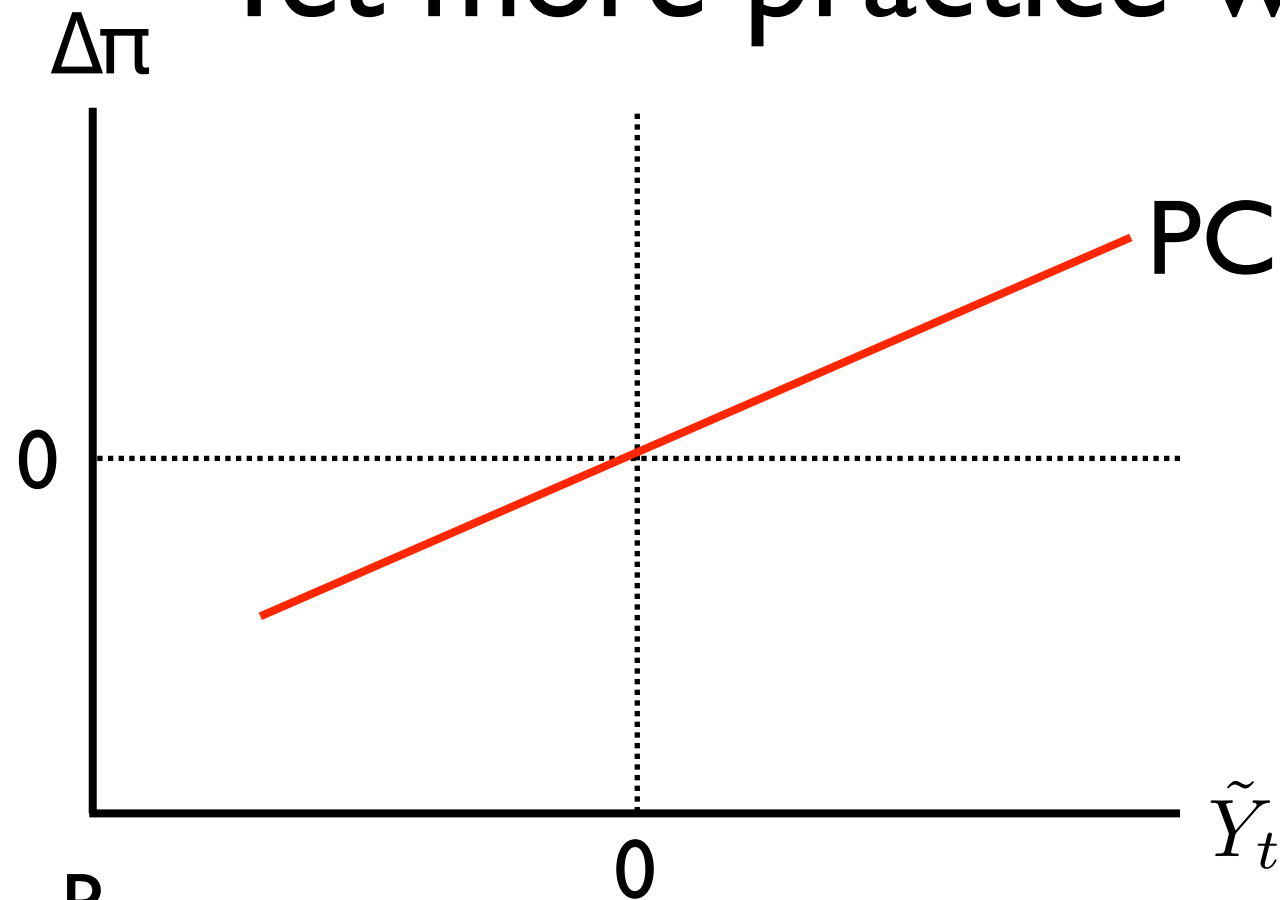


Q: Suppose the government expands its spending temporarily in order to stimulate the economy. What happens in the IS-MP or PC diagrams?

- What should the Fed do if it wants to pull inflation back down toward π_0 ?
- Raise the interest rate R and cause a recession
- Inflation will fall, and if the Fed keeps R high sufficiently long, π falls all the way back to π_0

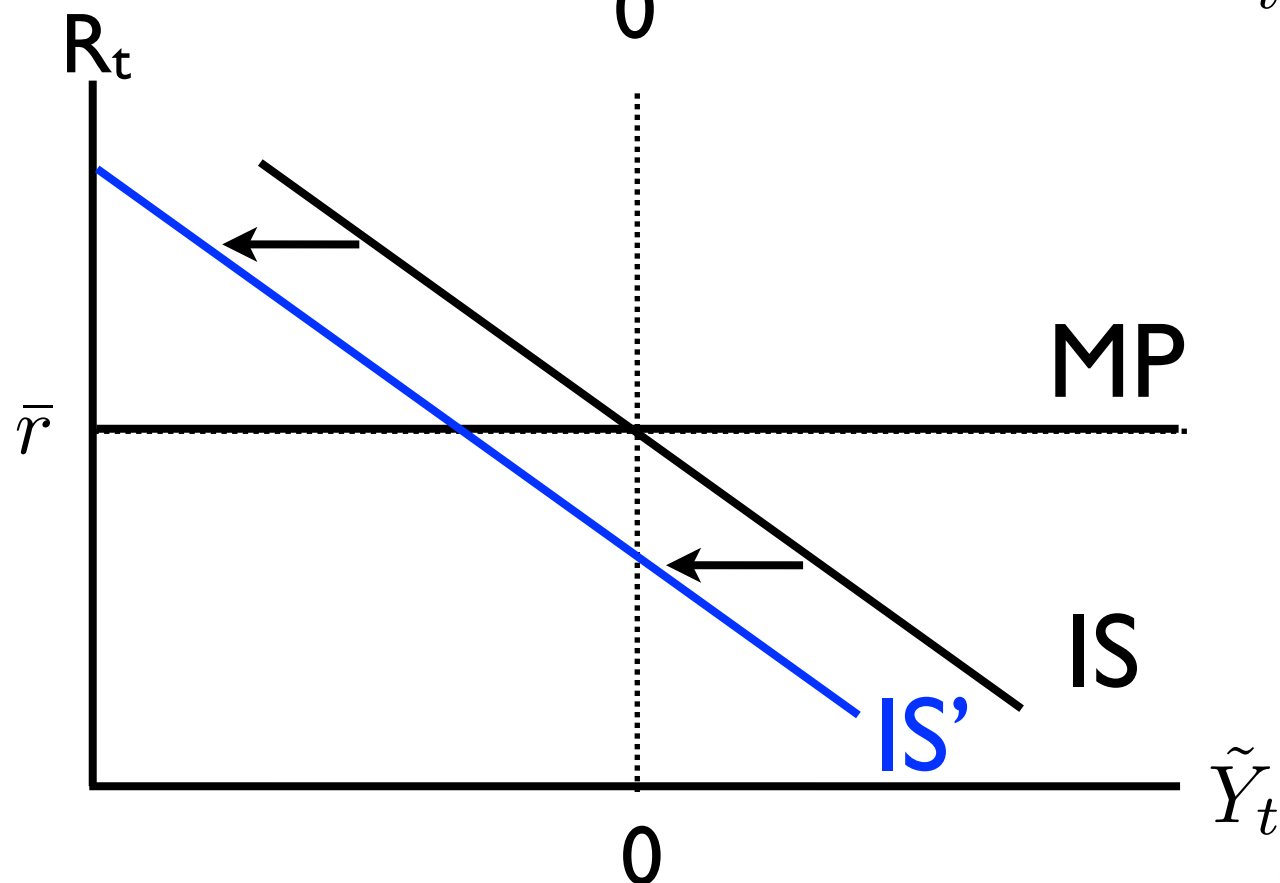


Yet more practice with IS-MP & PC (1/3)



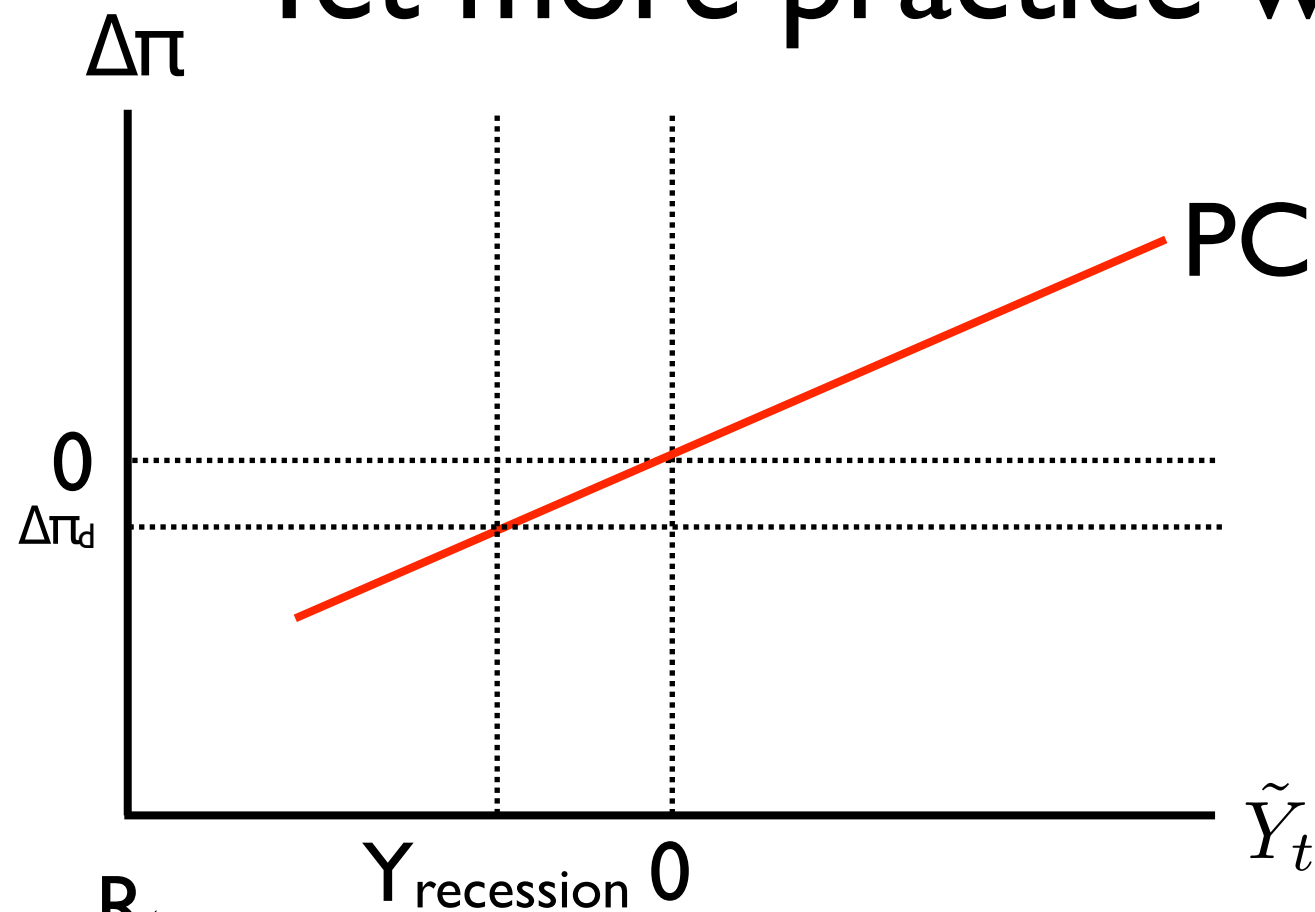
Q: Suppose consumers feel their houses, a major part of their wealth, have lost a lot of value, so they temporarily cut back on spending. What happens in the IS-MP or PC diagrams?

$$\tilde{Y}_t = \bar{a} - \bar{b} (R_t - \bar{r})$$



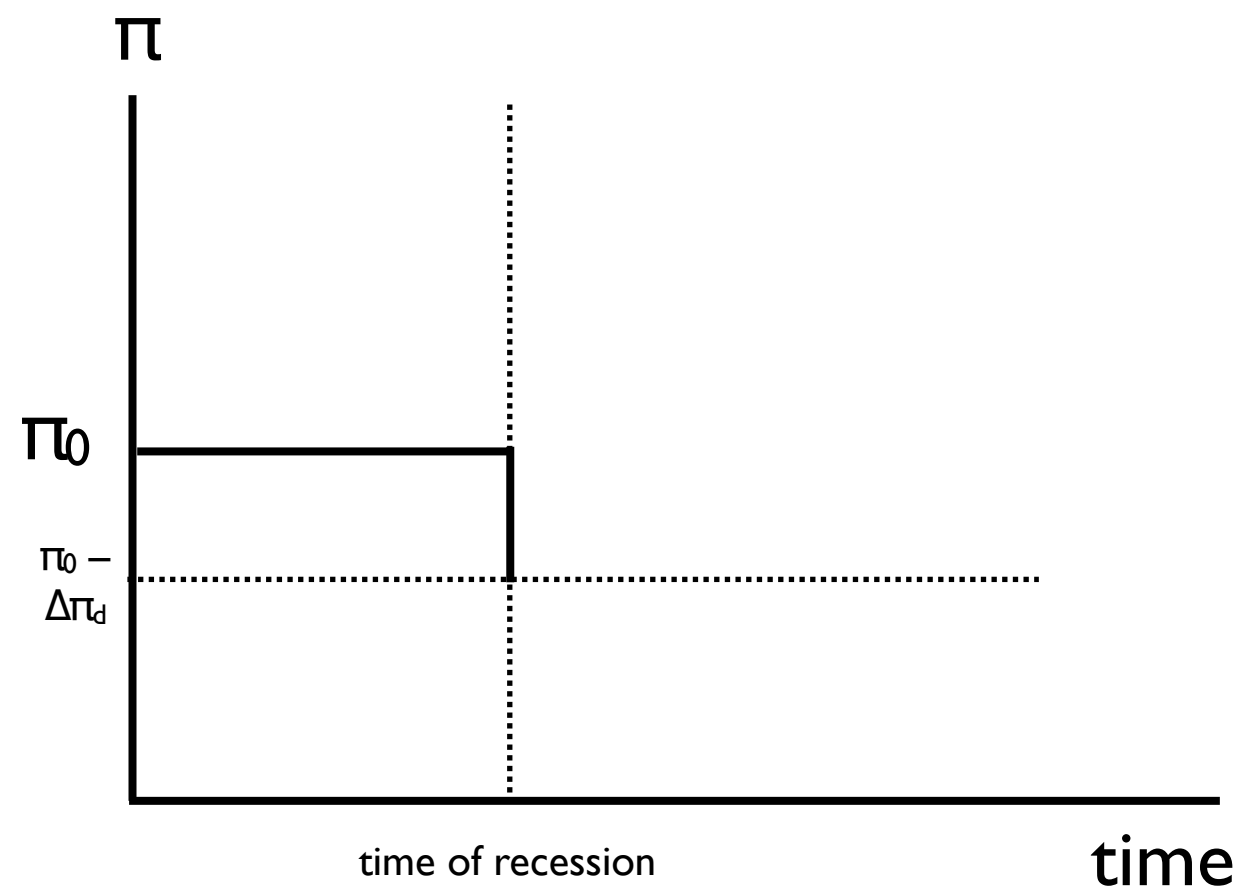
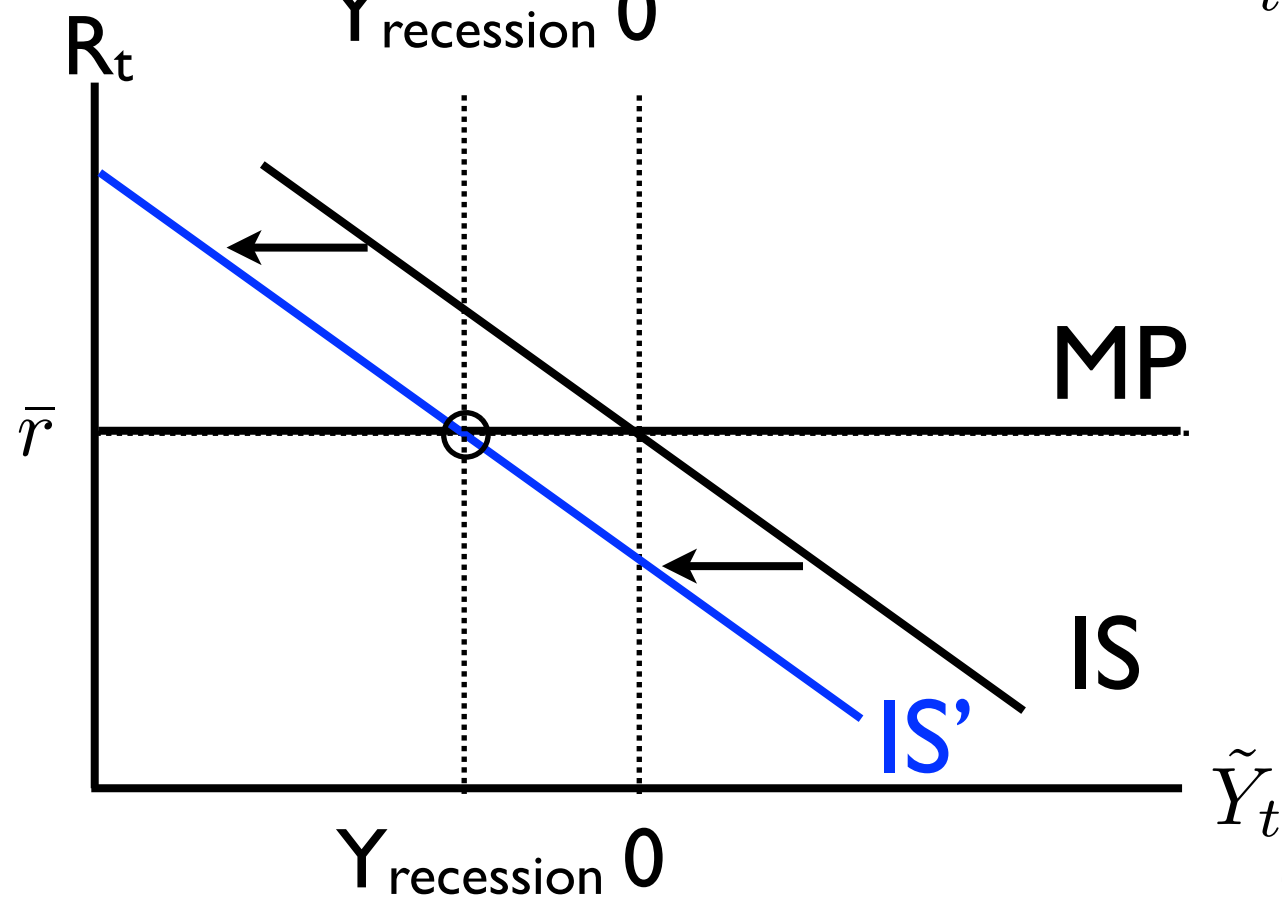
- This is a story about consumption spending that temporarily falls below its usual share of permanent income $\bar{a}_C = 0.64$
- When $\bar{a}_C$ falls, it pulls $\bar{a}$, the intercept of the IS curve, down with it.
- This is a *negative* aggregate demand shock of an amount equal to the reduction in $\bar{a}_C$
- It's a **leftward shift** in the IS Curve to IS'

Yet more practice with IS-MP & PC (2/3)

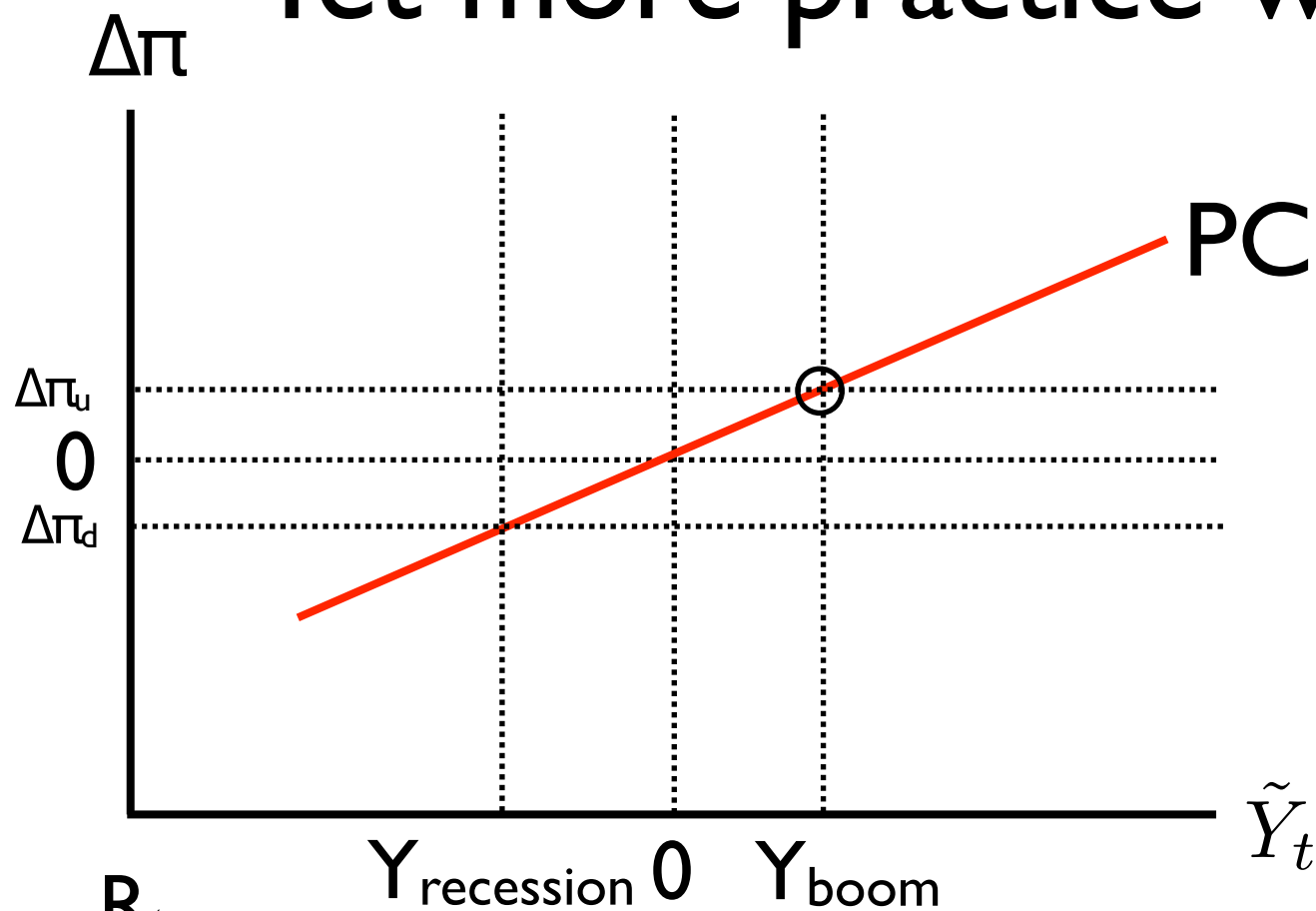


Q: Suppose consumers feel their houses, a major part of their wealth, have lost a lot of value, so they temporarily cut back on spending. What happens in the IS-MP or PC diagrams?

- The IS Curve has shifted to the left
- This produces a recession!
- This means that unless the Fed lowers the real interest rate R , inflation will decrease!

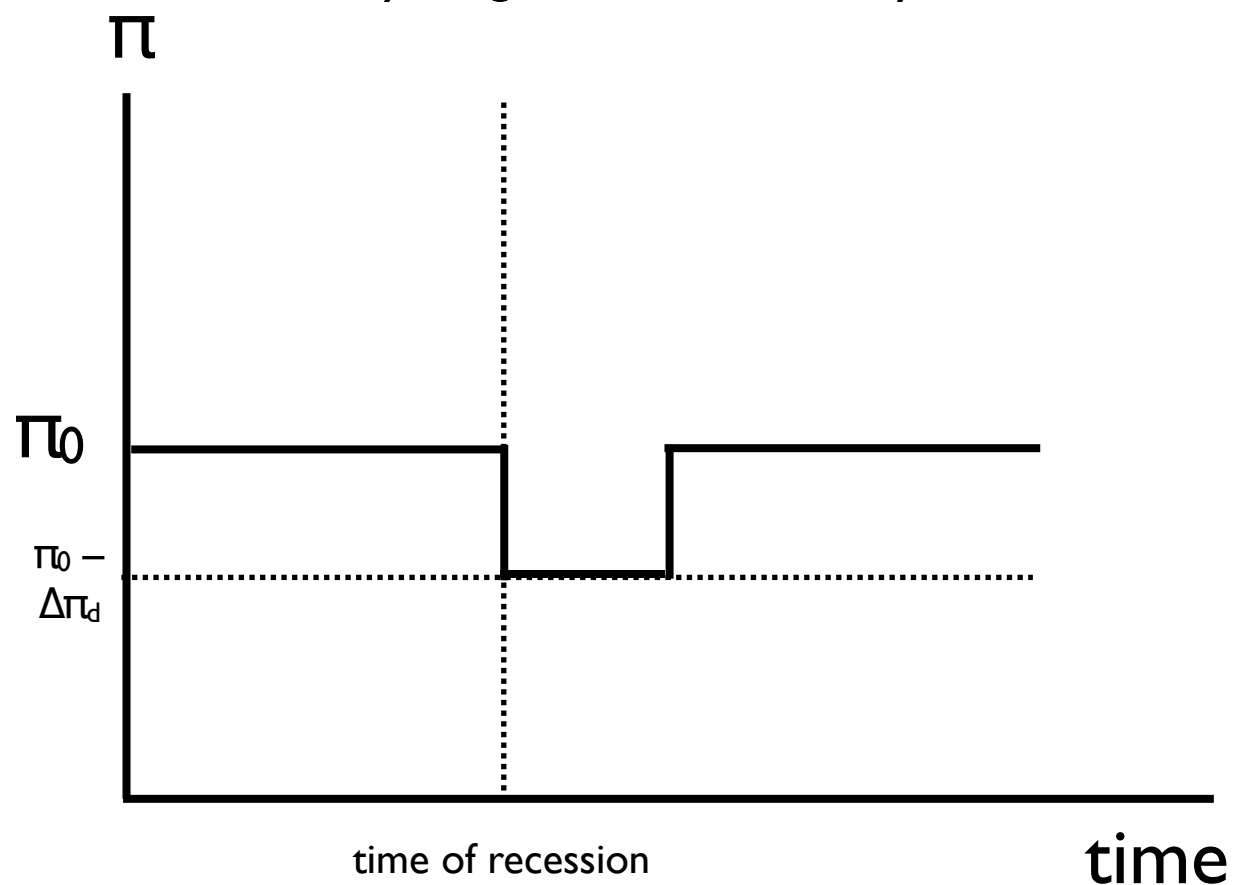
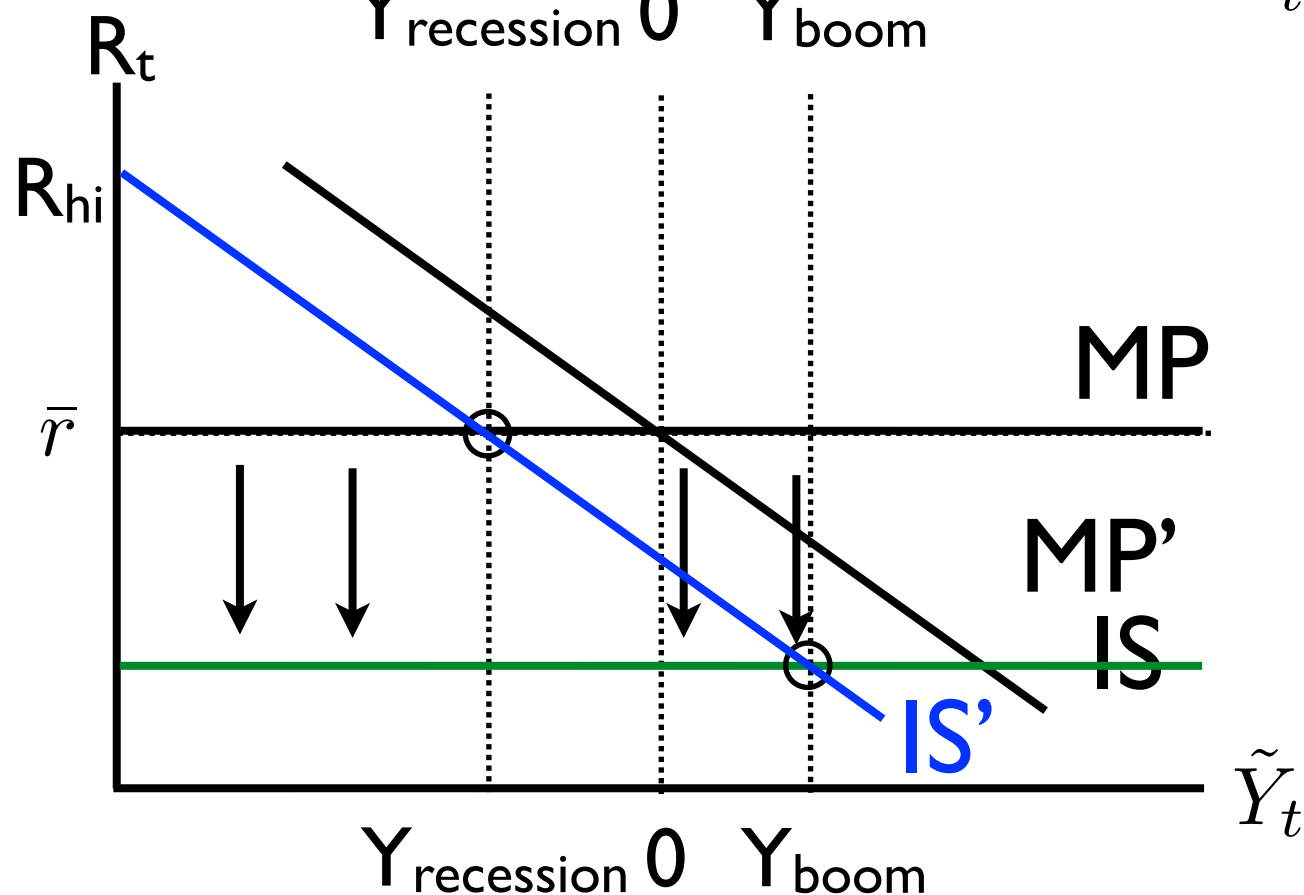


Yet more practice with IS-MP & PC (3/3)



Q: Suppose consumers feel their houses, a major part of their wealth, have lost a lot of value, so they temporarily cut back on spending. What happens in the IS-MP or PC diagrams?

- What should the Fed do if it wants to pull inflation back up toward π_0 ?
- Lower the interest rate R and *cause a boom*
- Inflation will rise, and if the Fed keeps R low for sufficiently long, π rises all the way back to π_0



How to think through the short-run model

- We care about
 - **Unemployment** and recessions: $\tilde{Y} < 0$
 - Inflation that is too high: $\pi > 0.02$ with target $\bar{\pi} = 0.02$
- Demand or supply considerations can affect either or both of these
- Exam or essay questions in Econ 100B are going to start with something about changes in demand or supply
- Then your task is to choose the right graph — either the IS-MP diagram or the Phillips curve PC diagram — and start modeling
 - What happens first
 - How the Fed reacts with monetary policy to try and obtain and maintain $\pi = \bar{\pi} = 0.02$

Why set monetary policy rules



Odysseus, bound to the mast of his ship, listens to the deadly song of the Sirens. British Museum, London [1843, 1103.31](#) 480-470 BCE

Homer and game theory

- Ben Bernanke, 2003, NABE conference, [“A Perspective on Inflation Targeting”](#)
- “I agree that human judgment should always be the ultimate source of policy decisions and that no central bank should — or is even able to — commit irrevocably in advance to actions that may turn out to be highly undesirable.
- “However, the intuition that more flexibility is always better than less flexibility is quite fallacious, a point understood long ago by Homer, who told of how Ulysses had himself tied to the mast so as not to fall victim to the songs of the Sirens.
- “More recently, the notion that more flexibility is always preferable has been pretty well gutted by modern game theory (not to mention modern monetary economics), which has shown in many contexts that the **ability to commit in advance often yields better outcomes.**”



Monetary policy rules

- A simple one in the textbook:

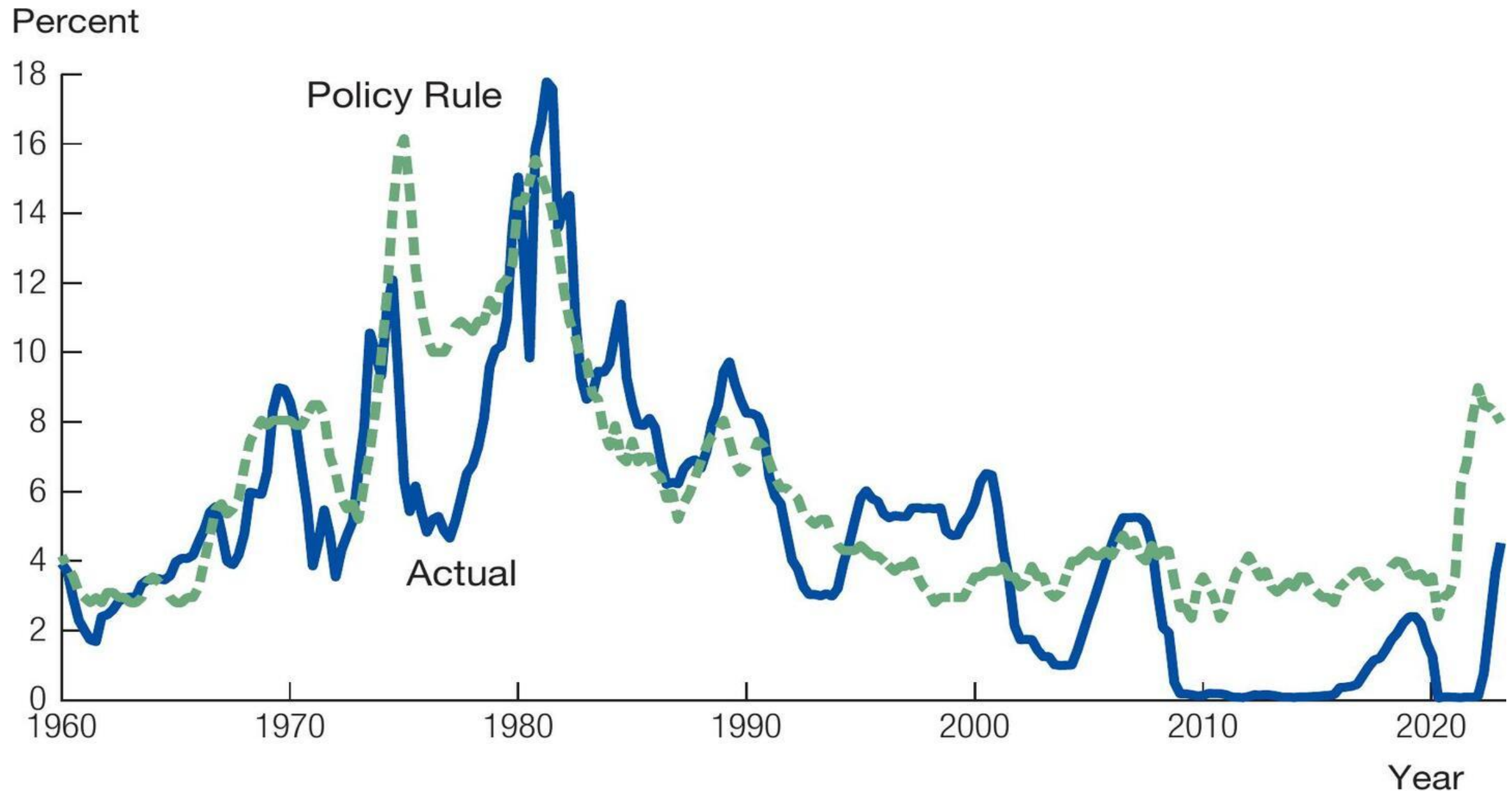
$$R - \bar{r} = \bar{m} (\pi - \bar{\pi}) \quad \text{with } \bar{m} = \frac{1}{2} \text{ and } \bar{\pi} = 0.02$$

- In words:
 - Raise R above the marginal product of capital, $\bar{r}$, when inflation is above 2%
 - Lower R below the marginal product of capital, $\bar{r}$, when inflation is below 2%
- Inspired by the **Taylor rule** suggested by Stanford economist [John B. Taylor](#) in the 1990s. His also included a $\tilde{Y}$ term

Why use a Taylor rule? Why not?

- It outlines precisely how the central bank will respond to inflation (and recessions)
- It ties the hands of the central bank, which promises that it will not respond otherwise
- It helps anchor expectations about the central bank's actions, and about inflation
- **An inflation target**
 - Can accomplish similar things
 - Avoids a problem in modern monetary policy: the **zero lower bound**

The textbook's graphic does not show the zero lower bound ever being breached



Source: The FRED database.

Shows a Taylor rule with $\bar{m} = 0.5$, $\bar{r} = 0.02$, and $\bar{\pi} = 0.02$

But [Nakamura, Riblier, and Steinsson \(2025\)](#) show the zero lower bound problem

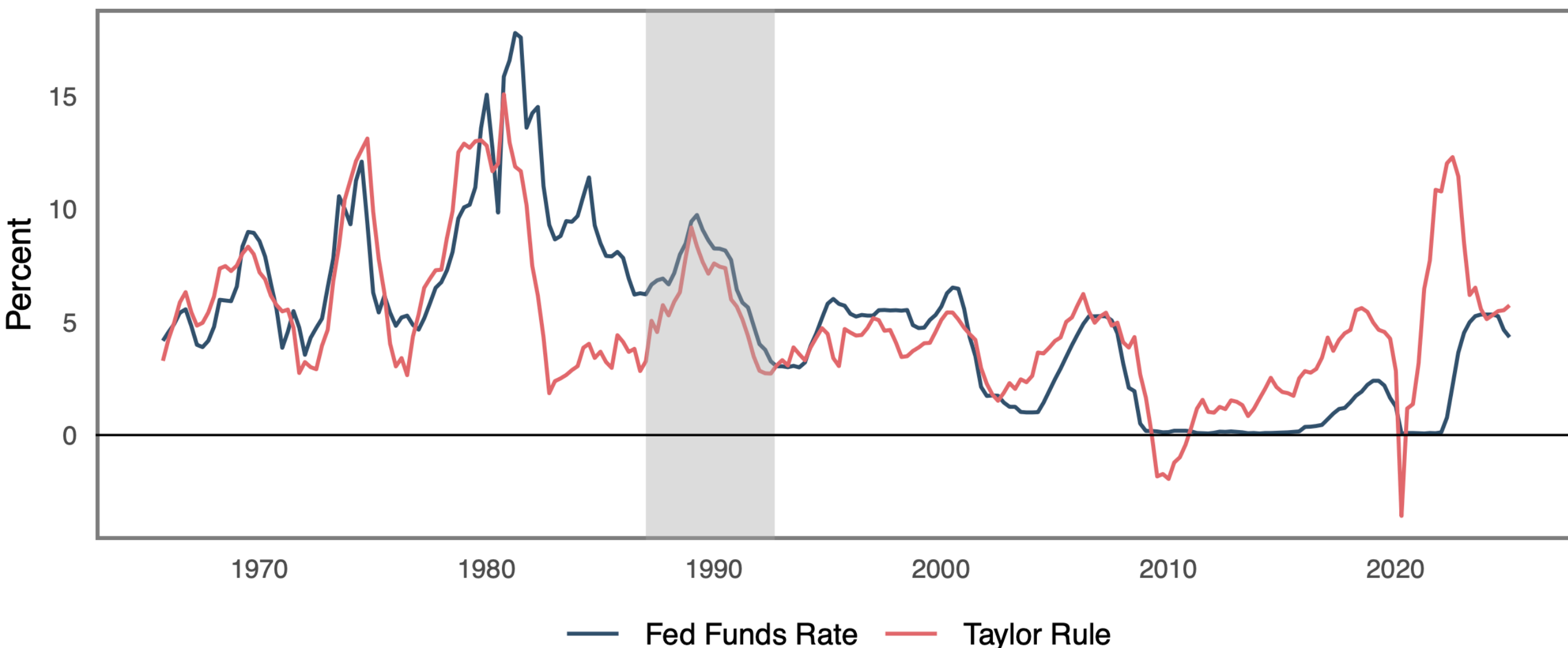


Figure 3: Original Taylor Rule with Real-Time Data

Note: The (light) red line shows the policy rate implied by the original Taylor rule—equation (1). We use real-time data for the GDP deflator and the Greenbook’s measure of the output gap (CBO output gap for last few years). The sample period is 1965Q4-2025Q1. The shaded area shows the sample period considered in [Taylor \(1993\)](#).

Shows a Taylor rule with $\bar{m} = 0.5$, $\bar{r} = 0.02$, and $\bar{\pi} = 0.02$